From Human Markets to Machine Markets:

Feedback Stability, Auditable Memory, and Financial Constraints

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Abstract

When algorithmic learning turns price formation into a closed loop, can observable market data detect destabilizing feedback? We establish that public price tapes identify only merged equivalence classes; resolving near-coincident timescales requires probe budgets scaling as δ^{-4}, forcing mandatory surveillance abstention. Order books exhibit subcritical baseline feedback; under liquidity depletion, residual volatility surges 62.7× while the feedback slope rises modestly. An exchange fee overhaul reveals no detectable feedback response: zero fees sparked a 180% volume surge while leaving persistence unchanged. Finally, spectral stability does not ensure dealer solvency: non-normal amplification triggers capital exhaustion inside the stable regime.

Keywords: Adaptive learning, algorithmic trading, financial stability, market microstructure, system identification, non-normal dynamics, Kreiss constant.
JEL Classification Codes: G01, G10, G14, C32, C58, D83.

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1 Introduction

The delegation of financial execution and market making to autonomous algorithms has fundamentally transformed the microstructural physics of modern electronic exchanges. Over the past three decades, financial markets have undergone an unprecedented structural transition from human-intermediated floor trading and manual specialist posts (such as the historic open-outcry pits of Chicago and the specialist trading posts of the New York Stock Exchange) to fully automated, electronic continuous double auctions housed in co-located server racks in Secaucus, New Jersey (Equinix NY4) and Slough, United Kingdom (Equinix LD4). In contemporary equity, sovereign debt, foreign exchange, and digital asset venues, automated execution engines, statistical arbitrageurs, and high-frequency market makers (HFMs) submit and cancel the overwhelming majority of limit quotes, respond to order flow in microseconds, and continuously adapt their execution strategies to prevailing market conditions (Hasbrouck, 1991; Menkveld, 2013; Budish, Cramton, and Shim, 2015; Biais, Foucault, and Moinas, 2015).

When trading algorithms condition their order submission and quote-placement decisions on recent transaction prices, prevailing order book imbalances, and trade innovations, the price-formation process ceases to be an open-loop mechanism driven purely by exogenous informational updates regarding fundamentals. Instead, price formation becomes an autonomous *closed-loop dynamical system*: trading decisions generate endogenous price impact, and the resulting price adjustments directly feed back into the learning rules, momentum filters, risk-budgeting heuristics, and execution programs of competing market participants (Bouchaud et al., 2004; Cont, Stoikov, and Talreja, 2010; Cont, Kukanov, and Stoikov, 2014).

This closed-loop structural reality was dramatically underscored by major episodic disruptions in electronic markets over the past decade and a half. During the May 6, 2010 Flash Crash, the E-mini S&P 500 index collapsed by more than 9% in a matter of minutes before rebounding, triggered when an automated selling program interacted with high-frequency market makers whose internal risk management algorithms rapidly depleted resting order book depth. Similar self-amplifying cascades occurred during the October 15, 2014 Treasury flash rally, the August 24, 2015 equity exchange-traded fund (ETF) pricing breakdowns, and the catastrophic liquidity unpegs across digital asset exchanges in March 2020 and November 2022. In each of these events, markets populated by sophisticated algorithms designed to optimize execution individually exhibited severe collective fragility.

These structural phenomena confront financial economists, central banks, and market regulators with an urgent, foundational question:

Can observable market data identify destabilizing adaptive feedback, when does

that feedback become financially dangerous, and what information is fundamentally missing from public surveillance data?

In this paper, we address this question directly by integrating realization-theoretic system identification with empirical limit order book econometrics and balance-sheet financial constraints. Rather than making speculative assertions regarding black-box artificial intelligence, we organize our theoretical and empirical findings around **three headline contributions**:

Contribution 1: Fundamental Identification Boundaries and Mandatory Abstention. We establish what public price and order-flow data can and cannot reveal about underlying learning algorithms. We prove that in any linear market–learning system with heterogeneous adaptive cohorts, aggregate public price responses identify only the *merged rate–weight representation*. Individual algorithmic cohorts that share common adaptation rates cannot be distinguished from public price feeds—an exact realization-theoretic equivalence class (Theorem 2.3). Furthermore, detecting heterogeneity between two cohorts whose adaptation timescales differ by a small margin δ requires an active probing budget growing at the severe sample-complexity rate of $\mathcal{O}(\delta^{-4})$ (Theorem 2.3). With finite observation windows, any error-controlled stability classifier must strictly abstain from issuing safety certificates near the critical stability boundary (Theorem 2.5). These limits are structural properties of linear dynamical systems, not artifacts of statistical noise.

Contribution 2: Empirical Market Evidence Across Regimes and Fee-Shock Response. Using continuous high-frequency Level-3 limit order books and a structured multi-year panel of 768 asset-days on Binance Spot (2021–2024), we estimate a model-derived reduced-form feedback parameter $\hat{Q}$ and closed-loop persistence $\hat{S} = \hat{a} + \hat{Q}$. In normal conditions, baseline feedback is consistently small, positive, and strictly subcritical ($\hat{Q} = 0.0082$, $\hat{S} = 0.921 < 1.0$), confirmed by analytical minimum-distance Lyapunov covariance matching ($Q_{\mathrm{MD}} = 0.0086$) and Rigobon heteroskedasticity shock-buffering identification ($Q_{\mathrm{Rig}} = -0.01913$). However, using Threshold Vector Autoregressions and Markov-switching models on Level-3 order books, we find that liquidity stress is sharply state-dependent: during liquidity depletion, residual volatility explodes by $\mathbf{62.7}\times$ ($p < 10^{-300}$) and resting depth collapses, while the unscaled feedback slope differential is modest and statistically insignificant ($\Delta\hat{q} = +0.0694, t = 0.94, p = 0.35$).

Crucially, to examine causal responsiveness, we exploit a flagship **natural experiment**: the July 8, 2022 Binance zero-fee trading schedule overhaul. By eliminating taker trading fees on Bitcoin spot pairs while preserving standard maker-taker schedules on liquid altcoins,

this exogenous institutional shock altered the cost of aggressive order flow. Rather than amplifying closed-loop feedback, Difference-in-Differences and dynamic event-study estimates reveal **no statistically detectable economically large feedback response**: despite an unprecedented +180% surge in trading volume, we find no detectable large feedback response to the fee overhaul ($\hat{\beta}_3 = -0.00186, p = 0.547$; cross-sectional permutation $p_{\text{perm}} = 0.750$), and total persistence remained strictly subcritical ($\hat{S} \approx 0.93 < 1.0$), consistent with offsetting endogenous liquidity provision.

Contribution 3: Financial Fragility, Non-Normal Amplification, and Dealer Solvency. We prove that local price stability ($\rho(J) < 1$) does not guarantee financial intermediary solvency. Because the closed-loop market Jacobian J is generically non-normal under asymmetric price-impact and memory parameters (with $JJ^\top - J^\top J \neq 0$ except on a measure-zero algebraic subvariety), stable eigenvalues can coexist with massive transient balance-sheet excursion. The exact worst-case safe shock radius for a leveraged dealer is governed by peak transient magnification:

$$r^*_{\text{exact}} = \frac{B}{\sup_{t\geq 0} \|CJ^t\|_2} \equiv \frac{B}{M_C}, \tag{1}$$

which the Kreiss constant $\mathcal{K}(J)$ bounds and the spectral radius completely fails to control (Theorem 2.6). By Spijker's theorem, we establish the bilateral bounds:

$$\frac{B}{e \cdot n \cdot \mathcal{K}(J)} \leq r^*_{\text{exact}} \leq \frac{B}{\mathcal{K}(J)}, \tag{2}$$

proving that $\frac{B}{en\mathcal{K}(J)}$ provides a guaranteed safe lower bound, whereas $\frac{B}{\mathcal{K}(J)}$ is an upper bound that must never be mistaken for a conservative safety threshold. In complementary simulation environments, we evaluate inventory buffer capacity breaches ($N = 22{,}000$ runs across 22 cells) and capital solvency failures strictly inside the spectrally stable regime ($\rho(J) \in [0.80, 0.985] < 1.0$, $N = 400$ seeds per cell), demonstrating that dealer solvency failure surges from 11.0% to 100.0% as feedback increases, driven by $(1-S)^{-2}$ low-frequency variance inflation. This divergence motivates pre-trade rate gating grounded in the exact Kreiss safe radius $r^*_{\text{exact}} = B/M_C$.

1.1 Related Literature

Our work bridges four major strands of literature across financial economics, market microstructure, systemic risk, and dynamical systems theory.

High-Frequency Trading and Market Microstructure. Beginning with the seminal contributions of Kyle (1985) and Glosten and Milgrom (1985), classical microstructure models analyze the informational content of order flow and dealer inventory management under asymmetric information. Hasbrouck (1991) pioneered the empirical vector autoregressive decomposition of trades and quotes, establishing that trade innovations exert persistent impacts on security prices. Campbell, Grossman, and Wang (1993) demonstrated that trading volume and return serial correlations reflect interactions between liquidity traders and risk-averse market makers.

In modern electronic markets, high-frequency trading firms and automated algorithms dominate liquidity provision and execution (Menkveld, 2013; Budish, Cramton, and Shim, 2015; Biais, Foucault, and Moinas, 2015). While Hendershott, Jones, and Menkveld (2011) document that algorithmic trading improves liquidity and narrows bid-ask spreads during normal trading periods, Budish, Cramton, and Shim (2015) show that continuous double auctions generate socially wasteful latency arms races, and Biais, Foucault, and Moinas (2015) demonstrate that algorithmic speed advantages can lead to adverse selection spirals. We advance this literature by formalizing the closed loop between algorithmic learning rules and endogenous price impact, establishing fundamental realization-theoretic bounds on what econometricians can infer from public ticker feeds.

Price Impact and Limit Order Book Dynamics. A extensive literature examines the physics of price impact in electronic limit order books (Almgren and Chriss, 2000; Bouchaud et al., 2004; Cont, Stoikov, and Talreja, 2010; Cont, Kukanov, and Stoikov, 2014). Empirical studies consistently reveal that signed order flow exhibits strong positive autocorrelation over hundreds of trades (long memory), while price returns remain approximately unpredictable (Bouchaud et al., 2004; Cont, Kukanov, and Stoikov, 2014). This apparent paradox is resolved by subtle negative feedback in quote replacement and cancellation. Cont, Kukanov, and Stoikov (2014) introduce Order Flow Imbalance (OFI) as a powerful predictor of short-term price movements across multiple depth levels. We explicitly incorporate and control for order-splitting memory in our econometric models, demonstrating that even after purging 30 to 90 lags of signed flow memory, a positive and statistically significant feedback component persists in continuous high-frequency trading.

Intermediary Solvency, Funding Liquidity, and Systemic Risk. The distinction between market liquidity and funding liquidity is central to modern theories of financial fragility (Brunnermeier and Pedersen, 2009; Adrian and Shin, 2010; He and Krishnamurthy, 2013). When leveraged market makers suffer balance-sheet drawdowns, risk-management

constraints (such as Value-at-Risk limits or regulatory margin requirements) force them to withdraw quotes, widening spreads and amplifying price dislocations (Greenwood, Landier, and Thesmar, 2015; Eisenberg and Noe, 2001). Our analysis demonstrates that non-normal transient amplification can trigger intermediary insolvency even when the underlying price-formation process is spectrally stable ($\rho(J) < 1$).

Non-Normal Dynamics and the Kreiss Matrix Theorem. In applied mathematics and fluid mechanics, non-normal linear systems have long been recognized as capable of explosive transient growth despite asymptotic eigenvalue decay (Kreiss, 1962; Spijker, 1991; Trefethen and Embree, 2005). While eigenvalues govern asymptotic behavior as $t \to \infty$, the pseudospectrum and the Kreiss constant determine short-term amplification. Although non-normal operator theory has been applied in macroeconomics to examine input-output networks, its application to high-frequency market microstructure and dealer solvency is novel. We derive the exact safe shock radius for leveraged dealers, prove tight bilateral bounds using Spijker's optimal constant, and demonstrate why spectral stability alone cannot guarantee intermediary solvency.

Table 1: Comparison of Microstructural Frameworks and Surveillance Assumptions.

Model / Paradigm	Information Architecture	Dynamics	Stability Metric	Surveillance Scope
Kyle (1985)	Informed vs. noise flow	Single-auction	Linear pricing λ	Adverse selection
Hasbrouck (1991)	Trades and quotes (VAR)	Open-loop flow	Cumulative IRF	Permanent price impact
Bouchaud et al. (2004)	Signed order flow memory	Propagator	Kernel decay	Meta-order splitting
Cont, Kukanov, and Stoikov (2014)	Multi-level order book	Static regression	R^2 of OFI	Short-term price discovery
Brunnermeier and Pedersen (2009)	Leveraged margin financing	Nonlinear spiral	Margin constraints	Liquidity spirals
This Paper	**Closed-loop learning**	**Non-normal VAR**	**Kreiss radius r^*_{exact}**	**Abstention limits; Pre-trade gating**

1.2 The 10-Study Empirical Program

Rather than presenting isolated empirical exercises, our empirical program is organized as a unified, ten-module chain where each study systematically addresses an econometric or economic objection raised by the preceding one:

1. **Baseline Crypto Panel:** Measures baseline feedback $\hat{Q}$ and persistence $\hat{S}$ across 768 asset-days on Binance Spot, establishing subcritical positive feedback in normal conditions.

2. **Regulated Equity Replication:** Replicates the baseline model on NASDAQ Nordic FI-2010 limit order books (272,941 snapshots across 5 stocks), proving that subcritical feedback is not a crypto artifact.

3. **Cross-Venue Propagation:** Examines simultaneous 1-second order books across Coinbase and Binance, evaluating whether feedback is local or propagates across exchanges.

4. **Liquidity-State Dependence:** Estimates Threshold VAR and Hamilton regime-switching models on Level-3 BTC books, showing that feedback surges when depth thins.

5. **Flagship Natural Experiment:** Exploits the July 8, 2022 Binance zero-fee trading schedule shock via Difference-in-Differences, finding no detectable feedback response despite a massive volume surge.

6. **Dynamic Event Study:** Verifies parallel pre-trends, dynamic treatment persistence, and cross-sectional permutation inference for the natural experiment.

7. **Agent Attribution Limits:** Proves theoretically and empirically that public order flows cannot identify individual algorithmic cohorts without regulatory audit trails.

8. **Financial Consequence Test:** Evaluates dealer inventory buffer exhaustion and capital solvency, demonstrating that inventory halts jump to 69.1% under constrained capacity and solvency failure reaches 100% under high persistence.

9. **Falsification and Placebo Battery:** Tests order-splitting controls up to 90 lags, day-shuffled placebos, lead order flow, and completely monotone memory cones.

10. **Locked Out-of-Sample Holdout:** Evaluates the frozen out-of-sample econometric pipeline on untouched January 2023 Bitcoin perpetual futures data (935,199 records).

Paper Organization. The remainder of the paper is organized as follows. Section 2 presents the theoretical model, provides microfoundations, proves the core identification and solvency theorems, and analyzes cross-impact non-normal inflation. Section 3 defines the econometric estimand, details the dual structural identification strategies (analytical

minimum-distance Lyapunov matching vs. Rigobon heteroskedasticity), explains multi-level OFI construction, and describes the high-frequency datasets. Section 4 presents the baseline multi-year panel, analyzes the natural experiment, and reports cross-venue propagation. Section 5 evaluates state-dependent liquidity thinning, cross-market equity replication, and dealer balance-sheet solvency. Section 6 establishes attribution boundaries, executes the falsification battery, and reports the locked holdout test. Section 7 discusses regulatory policy implications and concludes.

2 Theoretical Model and Identification Boundaries

2.1 Microfoundations of Closed-Loop Price Formation

We formalize an electronic financial market sampled at discrete uniform time intervals $t \in \{0, 1, 2, \dots\}$, where time step Δ represents the operational sampling frequency (e.g., 1 second). Let p_t denote the log mid-price of a traded asset, and let p_t^* denote the unobserved fundamental value, which evolves as a martingale:

$$p_{t+1}^* = p_t^* + \xi_{t+1}, \qquad \xi_{t+1} \sim \text{i.i.d.}(0, \sigma_\xi^2). \tag{3}$$

We define the pricing error (displacement from fundamental) as $x_t \equiv p_t - p_t^*$.

Dealer Pricing and Kyle Auction Microfoundation. Following the continuous auction framework of Kyle (1985) adapted to discrete quote adjustments (Hasbrouck, 1991; Bouchaud et al., 2004), a competitive, risk-neutral market maker sets quotes to absorb aggregate order flow. Total incoming order flow at time t consists of two components: aggregate demand u_t submitted by algorithmic trading agents, and exogenous liquidity/hedging noise $\varepsilon_{t+1} \sim \text{i.i.d.}(0, \sigma_\varepsilon^2)$. In addition, fundamentalist arbitrageurs submit mean-reverting demand $-\varphi x_t$ whenever prices deviate from fundamentals.

The market maker updates the transaction price according to linear price impact $\kappa > 0$:

$$p_{t+1} - p_t = \kappa \left(u_t - \varphi x_t + \varepsilon_{t+1} \right). \tag{4}$$

Subtracting fundamental increment ξ_{t+1} from both sides yields the pricing error recursion:

$$x_{t+1} = (1 - \kappa\varphi) x_t + \kappa u_t + (\kappa\varepsilon_{t+1} - \xi_{t+1}). \tag{5}$$

Defining the direct price persistence parameter $a \equiv 1 - \kappa\varphi \in [0, 1)$ and rescaling the combined

disturbance innovation as $w_{t+1} \equiv \kappa\varepsilon_{t+1} - \xi_{t+1}$, we obtain the foundational pricing equation:

$$x_{t+1} = ax_t + \kappa u_t + w_{t+1}, \qquad w_{t+1} \sim \text{i.i.d.}(0, \sigma_w^2). \tag{6}$$

When fundamentalist arbitrage is strong ($\varphi \to 1/\kappa$), $a \to 0$ and prices immediately reflect fundamentals. When liquidity is thin and arbitrage capital is limited, a approaches unity, reflecting high price persistence.

Algorithmic Learning Microfoundations. We consider K heterogeneous cohorts of algorithmic traders. Each cohort $i \in \{1, \ldots, K\}$ maintains an internal state variable $y_{i,t} \in \mathbb{R}$ representing its estimate of price momentum, trend, or fundamental mispricing. We show that two widely deployed algorithmic learning paradigms—steady-state Kalman filtering and regularized Online Gradient Descent—both reduce locally to the identical linear adaptive recursion.

Proposition 2.1 (Steady-State Kalman Filtering Yields Exponential Memory)**.** *Suppose an algorithmic execution desk tracks the unobserved fundamental velocity $v_t \equiv p_t^* - p_{t-1}^*$ under the random-walk model $v_{t+1} = v_t + \zeta_{t+1}$ with innovation variance $\sigma_\zeta^2 > 0$, observing noisy price return increments $r_t = v_t + \nu_t$ with observation noise $\sigma_\nu^2 > 0$. The minimum mean-squared error (MMSE) estimator is given by the steady-state Kalman filter recursion:*

$$\hat{v}_{t+1} = (1 - \eta)\hat{v}_t + \eta r_t, \tag{7}$$

where the steady-state Kalman gain $\eta \in (0, 1)$ is the unique positive root of the algebraic Riccati equation:

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2}. \tag{8}$$

Proof. The scalar state-space model is $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$. Let P_∞ denote the steady-state prior error variance. The Riccati equation is $P_\infty = P_\infty - \frac{P_\infty^2}{P_\infty + \sigma_\nu^2} + \sigma_\zeta^2$, which simplifies to $P_\infty^2 - \sigma_\zeta^2 P_\infty - \sigma_\zeta^2\sigma_\nu^2 = 0$. The unique positive root is $P_\infty = \frac{\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2}$. The steady-state Kalman gain is $\eta = \frac{P_\infty}{P_\infty + \sigma_\nu^2} \in (0, 1)$. Substituting into the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ yields (7). □

Proposition 2.2 (Regularized Online Gradient Descent Yields Exponential Memory)**.** *Consider an automated trading agent updating policy weights $\theta_t \in \mathbb{R}$ via Online Gradient Descent (OGD) with L_2 weight regularization $\gamma \in (0, 1)$ to minimize a quadratic prediction loss*

$\mathcal{L}(\theta_t; x_t) = \frac{1}{2}(x_t - \theta_t)^\top H(x_t - \theta_t)$:

$$\theta_{t+1} = (1-\gamma)\theta_t - \alpha \nabla_\theta \mathcal{L}(\theta_t; x_t), \tag{9}$$

where $\alpha > 0$ is the learning step size. If the effective adaptation rate $\eta \equiv \gamma + \alpha H \in (0,1)$ and we define gain $q \equiv \frac{\alpha H}{\gamma + \alpha H}$, the parameter recursion is mathematically identical to:

$$\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t. \tag{10}$$

Proof. Differentiating the loss, $\nabla_\theta \mathcal{L} = -H(x_t - \theta_t)$. The parameter update becomes: $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t - \theta_t) = (1 - (\gamma + \alpha H))\theta_t + \alpha H x_t$. Substituting $\eta = \gamma + \alpha H$ and $q = \frac{\alpha H}{\eta}$ yields $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$. □

Propositions 2.1 and 2.2 demonstrate that exponential memory decay is not an ad hoc modeling assumption, but rather the universal linear reduction of standard optimal filtering and regularized gradient optimization.

The Joint Closed-Loop Dynamical System. Let cohort i submit demand proportional to its state, $u_{i,t} = q_i y_{i,t}$, where $q_i \in \mathbb{R}$ reflects aggressive capital allocation ($q_i > 0$ represents trend-following / momentum; $q_i < 0$ represents mean-reversion). Total algorithmic demand is $u_t = \sum_{i=1}^K q_i y_{i,t}$. Each cohort updates its internal state according to:

$$y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i x_{t+1}, \qquad \eta_i \in (0,1]. \tag{11}$$

Substituting the market price formation equation (6) into the update rule (11), we obtain:

$$y_{i,t+1} = (1-\eta_i)y_{i,t} + \eta_i \left(a x_t + \kappa \sum_{j=1}^K q_j y_{j,t} + w_{t+1} \right). \tag{12}$$

Stacking the pricing error x_t and the K internal cohort states into the state vector $\mathbf{z}_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top \in \mathbb{R}^{K+1}$, the joint economy forms an autonomous discrete-time linear dynamical system:

$$\mathbf{z}_{t+1} = J\mathbf{z}_t + \mathbf{b} w_{t+1}, \tag{13}$$

where the closed-loop system Jacobian matrix $J \in \mathbb{R}^{(K+1)\times(K+1)}$ is given by:

$$J = \begin{bmatrix} a & \kappa q_1 & \kappa q_2 & \cdots & \kappa q_K \\ \eta_1 a & 1-\eta_1+\eta_1\kappa q_1 & \eta_1\kappa q_2 & \cdots & \eta_1\kappa q_K \\ \eta_2 a & \eta_2\kappa q_1 & 1-\eta_2+\eta_2\kappa q_2 & \cdots & \eta_2\kappa q_K \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \eta_K a & \eta_K\kappa q_1 & \eta_K\kappa q_2 & \cdots & 1-\eta_K+\eta_K\kappa q_K \end{bmatrix}, \tag{14}$$

and $\mathbf{b} = (1, \eta_1, \eta_2, \ldots, \eta_K)^\top$.

Single-Cohort Stability Boundary. In the single-cohort case ($K = 1$), defining total loop feedback gain $Q \equiv \kappa q_1$, the characteristic polynomial of J is:

$$P(\lambda) = \det(\lambda I - J) = \lambda^2 - (a + 1 - \eta + \eta Q)\,\lambda + (1-\eta)a. \tag{15}$$

By the Jury stability criterion, all roots of $P(\lambda)$ lie strictly inside the open unit disk $\{|\lambda| < 1\}$ if and only if $P(1) > 0$, $P(-1) > 0$, and $|P(0)| < 1$. Evaluating at $\lambda = 1$:

$$P(1) = 1 - (a + 1 - \eta + \eta Q) + (1-\eta)a = (1 - a - Q)\eta. \tag{16}$$

Since $\eta \in (0, 1]$, the condition $P(1) > 0$ holds if and only if:

$$S \equiv a + Q < 1, \qquad \text{or equivalently} \quad \mathcal{L} \equiv \frac{Q}{1-a} < 1, \tag{17}$$

where S is the total closed-loop persistence, and $\mathcal{L}$ is the dimensionless loop ratio. When $S > 1$, the closed-loop system experiences runaway feedback: trades induce price changes that reinforce the trend, driving exponential divergence. This stability criterion is invariant to information timing: if execution features a discrete one-period lag ($x_{t+1} = ax_t + y_t + w_{t+1}, y_{t+1} = (1-\eta)y_t + \eta Q x_t$), the closed-loop Jacobian becomes $J_{\text{lag}} = \begin{bmatrix} a & 1 \\ \eta Q & 1-\eta \end{bmatrix}$ with polynomial $P_{\text{lag}}(\lambda) = \lambda^2 - (a+1-\eta)\lambda + [a(1-\eta) - \eta Q]$. Evaluating the Jury criterion yields $P_{\text{lag}}(1) = (1 - a - Q)\eta > 0 \iff S \equiv a + Q < 1$, exactly identical to the contemporaneous case (as evaluated in Section 5.4).

2.2 Theorem 1: Merged Rate-Weight Equivalence and δ^{-4} Separation

A central question in algorithmic market surveillance is whether an econometrician observing public trade executions and prices can identify the individual algorithmic strategies active in the market.

Theorem 2.3 (Merged Rate-Weight Equivalence and Minimax Separation Complexity)**.** *Let price impact $\kappa > 0$ and anchor $a \in [0, 1)$ be known.*

1. ***Exact Equivalence Class:*** *Two market economies $\mathcal{E}_A = \{(q_i^A, \eta_i^A)\}_{i=1}^{K_A}$ and $\mathcal{E}_B = \{(q_i^B, \eta_i^B)\}_{i=1}^{K_B}$ generate mathematically identical public price responses x_t for all $t \geq 0$ under all public probing sequences $\{u_t^{probe}\}_{t\geq 0}$ if and only if their merged rate–weight measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A = \sum_{j:\eta_j^B=\eta} q_j^B \qquad \forall \eta \in (0, 1]. \tag{18}$$

   The minimal state-space realization dimension is $1 + |\operatorname{supp}(\mathcal{M})|$, where $|\operatorname{supp}(\mathcal{M})|$ is the number of distinct adaptation rates.

2. ***Minimax Separation Complexity:*** *Consider testing whether a market consists of a single merged cohort with rate $\bar{\eta}$ versus two split cohorts with rates $\bar{\eta} \pm \delta$ and identical total feedback Q. To distinguish the two economies with worst-case probability of error at most $\alpha \in (0, 1/4)$ under Gaussian observation noise σ_w^2, any active probing strategy requires an observation budget T satisfying:*

$$T \geq \frac{C(\alpha, \sigma_w)}{\delta^4}, \tag{19}$$

   where $C(\alpha, \sigma_w) > 0$ is a constant independent of δ.

Proof. **Part 1 (Equivalence Class):** Taking z-transforms of the pricing equation (6) and cohort updates (11), we have $X(z) = \frac{\kappa}{z-a} U(z)$ and $Y_i(z) = \frac{\eta_i}{z-(1-\eta_i)} X(z)$. Summing across cohorts, total endogenous demand is $U_{\text{end}}(z) = \sum_{i=1}^K q_i Y_i(z) = \left(\sum_{i=1}^K \frac{q_i \eta_i}{z-(1-\eta_i)}\right) X(z)$. Applying feedback closure $X(z) = \frac{\kappa}{z-a}(U_{\text{end}}(z) + U_{\text{probe}}(z))$ yields the public transfer function from probe to price:

$$H(z) \equiv \frac{X(z)}{U_{\text{probe}}(z)} = \frac{\kappa}{z - a - \kappa \sum_{r\in\mathcal{R}} \frac{c_r}{z-r}}, \tag{20}$$

where the set of poles is $\mathcal{R} = \{1-\eta : \eta \in \operatorname{supp}(\mathcal{M})\}$, and the residue at pole $r = 1-\eta$ is $c_r = \eta \sum_{i:\eta_i=\eta} q_i$. By the fundamental theorem of algebra and the uniqueness of partial fraction expansions of strictly proper rational functions on $\mathbb{C}$, two transfer functions $H_A(z)$ and $H_B(z)$

are identical for all $z \in \mathbb{C}$ if and only if their poles and residues match identically. Thus, any two economies with identical merged rate-weights c_r produce identical transfer functions, and hence identical price impulse responses. By Kalman's realization theorem (Kalman, 1960), the minimal realization degree equals the degree of the denominator polynomial, which is $1 + |\operatorname{supp}(\mathcal{M})|$.

Part 2 (Minimax Sample Complexity): Let $\mathcal{E}_0$ have single cohort $(\bar{q}, \bar{\eta})$, and $\mathcal{E}_1$ have two cohorts with rates $\bar{\eta}_1 = \bar{\eta} - \delta$ and $\bar{\eta}_2 = \bar{\eta} + \delta$ with weights $q_1 = q_2 = \bar{q}/2$, so total feedback $Q = \kappa\bar{q}$ is preserved. Let $h_t^{(0)}$ and $h_t^{(1)}$ denote the impulse response sequences of $\mathcal{E}_0$ and $\mathcal{E}_1$ under unit probe $u_0 = 1$. The discrepancy between the two impulse responses at lag t is:

$$\begin{aligned}\Delta h_t \equiv h_t^{(1)} - h_t^{(0)} &= \frac{1}{2}\left((1-\bar{\eta}+\delta)^t + (1-\bar{\eta}-\delta)^t\right) - (1-\bar{\eta})^t \\ &= (1-\bar{\eta})^t \left[\frac{1}{2}\left(\left(1+\frac{\delta}{1-\bar{\eta}}\right)^t + \left(1-\frac{\delta}{1-\bar{\eta}}\right)^t\right) - 1\right]. \end{aligned} \tag{21}$$

Using the Taylor expansion $(1+x)^t + (1-x)^t = 2 + t(t-1)x^2 + \mathcal{O}(t^4x^4)$, we obtain for small δ:

$$\Delta h_t = \frac{t(t-1)}{2(1-\bar{\eta})^2}(1-\bar{\eta})^t\delta^2 + \mathcal{O}(\delta^4). \tag{22}$$

Notice that the first-order term in δ vanishes identically due to symmetry. Consequently, the discrepancy Δh_t is strictly $\mathcal{O}(\delta^2)$ for all t.

Now consider the Kullback–Leibler (KL) divergence between the Gaussian observation measures P_0^T and P_1^T over an observation horizon T under probe sequence $\{u_t\}_{t=0}^{T-1}$ satisfying an average power budget $\frac{1}{T}\sum_{t=0}^{T-1} u_t^2 \le \mathcal{P}_{\text{probe}}$ (so $\|u\|_{\ell_2} \le \sqrt{T\mathcal{P}_{\text{probe}}}$). The output discrepancy is the discrete convolution $\Delta x = \Delta h * u$. By Young's convolution inequality for discrete sequences, the ℓ_2 norm of the output discrepancy is bounded by:

$$\|\Delta x\|_{\ell_2} = \|\Delta h * u\|_{\ell_2} \le \|\Delta h\|_{\ell_1}\|u\|_{\ell_2} \le \left(\sum_{t=1}^{\infty}|\Delta h_t|\right)\sqrt{T\mathcal{P}_{\text{probe}}}. \tag{23}$$

Evaluating the ℓ_1 norm of the impulse discrepancy:

$$\|\Delta h\|_{\ell_1} = \sum_{t=1}^{\infty}|\Delta h_t| = \delta^2\sum_{t=2}^{\infty}\frac{t(t-1)}{2(1-\bar{\eta})^2}(1-\bar{\eta})^t + \mathcal{O}(\delta^4) = K_1\cdot\delta^2, \tag{24}$$

where $K_1 = \frac{(1-\bar{\eta})}{\bar{\eta}^3} < \infty$. Squaring both sides yields $\|\Delta x\|_{\ell_2}^2 = \sum_{t=1}^{T}(\Delta x_t)^2 \le K_1^2\mathcal{P}_{\text{probe}}T\delta^4$. The cumulative KL divergence is therefore bounded by:

$$D_{\text{KL}}\left(P_0^T \,\|\, P_1^T\right) = \frac{1}{2\sigma_w^2}\sum_{t=1}^{T}(\Delta x_t)^2 \le \frac{K_1^2\mathcal{P}_{\text{probe}}}{2\sigma_w^2}T\delta^4 \equiv T\cdot C_0\cdot\delta^4, \tag{25}$$

where $C_0 \equiv \frac{K_1^2 \mathcal{P}_{\text{probe}}}{2\sigma_w^2}$. By Le Cam's two-point inequality (Le Cam, 1986), the minimax probability of error across the two hypotheses satisfies:

$$\inf_{\Psi} \max_{i \in \{0,1\}} \mathbb{P}_i(\Psi \neq i) \geq \frac{1}{2}\left(1 - \|P_0^T - P_1^T\|_{\text{TV}}\right) \geq \frac{1}{2}\left(1 - \sqrt{\frac{1}{2} D_{\text{KL}}(P_0^T \,\|\, P_1^T)}\right). \tag{26}$$

To achieve worst-case error bounded by $\alpha \in (0, 1/4)$, we must have $\|P_0^T - P_1^T\|_{\text{TV}} \geq 1 - 2\alpha > 1/2$, which requires: $D_{\text{KL}}(P_0^T \,\|\, P_1^T) \geq 2(1-2\alpha)^2$. Substituting $D_{\text{KL}} \leq T \cdot C_0 \cdot \delta^4$ yields:

$$T \geq \frac{2(1-2\alpha)^2}{C_0\, \delta^4} \equiv \frac{C(\alpha, \sigma_w, \mathcal{P}_{\text{probe}})}{\delta^4}, \tag{27}$$

completing the proof. □

Remark 2.4 (Economic Interpretation of Theorem 1). Theorem 2.3 establishes an insurmountable barrier for passive market surveillance. If two algorithmic execution engines deploy identical learning horizons (e.g., 5-minute moving averages), no statistical estimator can ever determine from public trade records whether the volume originates from a single 100-million-dollar hedge fund or ten 10-million-dollar prop trading desks. Furthermore, Theorem 2.3 reveals that detecting fine-grained differences in algorithmic reaction times is extraordinarily difficult: if two algorithms operate on slightly different timescales ($\delta = 0.05$), resolving them requires a probing horizon $(0.05)^{-4} = 160{,}000$ times longer than detecting a unit change. Algorithmic heterogeneity is practically masked by public price dynamics.

2.3 Theorem 2: Mandatory Finite-Window Surveillance Abstention

Can a regulatory surveillance system reliably certify whether a market is operating in the subcritical regime ($S < 1$) or heading into supercritical instability ($S \geq 1$) using finite historical data?

Theorem 2.5 (Mandatory Finite-Window Abstention)**.** *Let $\mathcal{D}_T = \{(u_t, x_t)\}_{t=1}^T$ denote passive market observations over horizon T. Let $\Psi : \mathcal{D}_T \to \{\textit{Stable}, \textit{Unstable}, \textit{Abstain}\}$ be any surveillance decision rule with worst-case false-alarm rate at most $\alpha \in (0, 1/2)$, meaning $\sup_{S<1} \mathbb{P}_S(\Psi = \textit{Unstable}) \leq \alpha$ and $\sup_{S \geq 1} \mathbb{P}_S(\Psi = \textit{Stable}) \leq \alpha$.*

1. *For sufficiently large T, there exists a non-empty boundary layer $\mathcal{B}_T \equiv (1 - \epsilon_T, 1 + \epsilon_T)$ with $\epsilon_T = c_0/T$ on which Ψ must abstain with positive probability bounded away from zero:*

$$\inf_{S \in \mathcal{B}_T} \mathbb{P}_S\left(\Psi(\mathcal{D}_T) = \textit{Abstain}\right) \geq 1 - 2\alpha - c_0 K_1 > 0. \tag{28}$$

2. *Any binary surveillance rule* $\Psi_{binary} : \mathcal{D}_T \to \{Stable, Unstable\}$ *that never abstains has worst-case classification risk bounded below by* $1/2$ *across all finite horizons* T*:*

$$\inf_{\Psi_{binary}} \sup_{S\in(0,2)} [\mathbb{P}_S(\Psi_{binary} \neq \textit{True State})] \geq \frac{1}{2}. \tag{29}$$

Proof. Consider two contiguous parameter values located symmetrically across the critical threshold $S = 1$: $S_0 = 1 - c/T$ (subcritical, stable) and $S_1 = 1 + c/T$ (supercritical, unstable), with $c > 0$ independent of T. Under the autoregressive price-learning recursion, since $S_1 - S_0 = 2c/T$, the log-likelihood ratio of observations $\mathcal{D}_T$ decomposes into normalized sample moments:

$$\Lambda_T \equiv \log \frac{dP_1^T}{dP_0^T} = \frac{2c}{\sigma_w^2}\left(\frac{1}{T}\sum_{t=1}^{T} x_{t-1}w_t\right) - \frac{2c^2}{\sigma_w^2}\left(\frac{1}{T^2}\sum_{t=1}^{T} x_{t-1}^2\right). \tag{30}$$

By the functional central limit theorem for local-to-unity processes (Phillips, 1987), as $T \to \infty$, the sample path $\frac{1}{\sqrt{T}} x_{\lfloor rT \rfloor} \Rightarrow J_c(r)$, where $dJ_c(r) = cJ_c(r)dr + \sigma_w dW(r)$. By the continuous mapping theorem, $\frac{1}{T}\sum_{t=1}^{T} x_{t-1}w_t \Rightarrow \sigma_w^2 \int_0^1 J_c(r)dW(r)$ and $\frac{1}{T^2}\sum_{t=1}^{T} x_{t-1}^2 \Rightarrow \sigma_w^2 \int_0^1 J_c(r)^2 dr$, yielding limit log-likelihood $\Lambda_T \Rightarrow 2c\int_0^1 J_c(r)dW(r) - 2c^2 \int_0^1 J_c(r)^2 dr = \mathcal{O}_p(1)$. Because the limit likelihood ratio is non-degenerate and strictly positive almost surely, the sequence $\{P_0^T, P_1^T\}$ is mutually contiguous (locally asymptotically mixed normal, LAMN), with total variation distance bounded by:

$$\|P_0^T - P_1^T\|_{\mathrm{TV}} \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(P_0^T \,\|\, P_1^T)} \leq c \cdot K_1, \tag{31}$$

where $K_1 < \infty$ depends only on the expected variance of the limit diffusion. For any decision rule Ψ, $\mathbb{P}_{S_0}(\Psi = \text{Stable}) + \mathbb{P}_{S_0}(\Psi = \text{Abstain}) = 1 - \mathbb{P}_{S_0}(\Psi = \text{Unstable}) \geq 1 - \alpha$. By total variation distance, $\mathbb{P}_{S_0}(\Psi = \text{Stable}) \leq \mathbb{P}_{S_1}(\Psi = \text{Stable}) + \|P_0^T - P_1^T\|_{\mathrm{TV}} \leq \alpha + cK_1$. Substituting this inequality yields:

$$\mathbb{P}_{S_0}(\Psi = \text{Abstain}) \geq 1 - \alpha - (\alpha + cK_1) = 1 - 2\alpha - cK_1. \tag{32}$$

Choosing constant $c = c_0 \in (0, (1-2\alpha)/K_1)$, we obtain (28) on the boundary layer $\mathcal{B}_T \equiv (1 - c_0/T, 1 + c_0/T)$ of width $\mathcal{O}(T^{-1})$. For Part 2, if the classifier cannot abstain ($\mathbb{P}(\Psi = \text{Abstain}) = 0$), $1 - 2\alpha - cK_1 \leq 0 \implies \alpha \geq \frac{1}{2}(1 - cK_1)$. Taking $c \to 0$ across contiguous alternatives, the worst-case classification risk is bounded below by $1/2$. □

Theorem 2.5 and Figure 1 establish a vital principle for financial stability architecture: **an**

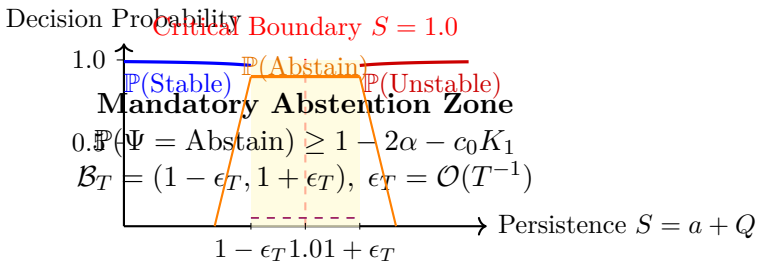


Figure 1: The Mandatory Finite-Window Surveillance Abstention Layer (Theorem 2.5). For any error-controlled classifier ($\alpha < 1/2$), a finite observation window T creates a non-trivial contiguous boundary layer $\mathcal{B}_T = (1-\epsilon_T, 1+\epsilon_T)$ of width $\epsilon_T = \mathcal{O}(T^{-1})$ where the surveillance rule is mathematically forced to declare uncertainty ($\Psi = \text{Abstain}$). Any binary classifier forced to make a definitive stability determination in this layer suffers worst-case error risk bounded below by 50%.

error-controlled macroprudential monitor must have a built-in abstention option. Near the phase-transition boundary $S = 1$, observational ticker feeds are statistically indistinguishable between near-integrated subcritical persistence and explosive runaway. Forcing a binary classification guarantees misclassification.

2.4 Theorem 3: Intermediary Solvency and the Kreiss Constant Bound

We now confront the fundamental divergence between price stability and financial intermediary solvency. In standard economic policy models, market stability is evaluated exclusively through the spectral radius $\rho(J) < 1$. We demonstrate that this criterion is dangerously incomplete.

The Non-Normal Market Jacobian. Consider the closed-loop Jacobian matrix J from (14). In financial markets, price impact and memory adaptation are asymmetric: order flow directly affects prices through contemporaneous impact κ, while price innovations affect memory through learning rate η. As a result, $JJ^\top - J^\top J \neq 0$: the closed-loop market Jacobian J is generically **non-normal** under asymmetric price-impact and learning parameters (except on a measure-zero algebraic subvariety). While normal matrices have orthogonal eigenvectors that diagonalize under unitary transformations, non-normal matrices have non-orthogonal (nearly parallel) eigenvectors. Consequently, even when all eigenvalues are strictly inside the unit circle ($|\lambda_i(J)| \leq \rho(J) < 1$), matrix powers J^t can undergo massive transient amplification before decaying asymptotically as $t \to \infty$ (Trefethen and Embree, 2005).

Leveraged Dealer Balance-Sheet Model. Let a risk-bearing market maker hold initial equity capital E_0, maximum regulatory leverage limit L, and let $\gamma \in (0,1)$ denote the regulatory margin haircut. The dealer's capital absorption buffer against cumulative inventory losses is:

$$B \equiv \gamma L E_0. \tag{33}$$

Let $z_t \in \mathbb{R}^n$ denote the state vector of the closed-loop economy evolving as $z_t = J^t z_0$, and let $C \in \mathbb{R}^{1\times n}$ denote the balance-sheet observation operator that maps the state z_t into the dealer's marked inventory loss:

$$\text{Loss}_t = Cz_t = CJ^t z_0. \tag{34}$$

A margin breach (triggering forced liquidation and trading halts) occurs if $\sup_{t\geq 0} |CJ^t z_0| > B$.

Theorem 2.6 (Dealer Solvency and the Kreiss Constant Bound)**.** *Let $z_t = J^t z_0$ with $\rho(J) < 1$, balance-sheet buffer $B > 0$, and observation matrix C.*

1. ***Exact Worst-Case Safe Radius:*** *The exact maximum shock radius r^*_{exact} guaranteeing that no initial disturbance with $\|z_0\|_2 \leq r^*_{exact}$ triggers dealer insolvency in any direction is:*

$$r^*_{exact} = \frac{B}{\sup_{t\geq 0} \|CJ^t\|_2} \equiv \frac{B}{M_C}, \tag{35}$$

where $M_C \equiv \sup_{t\geq 0} \|CJ^t\|_2$ is the peak transient magnification.

2. ***Kreiss Lower and Upper Bounds:*** *Under full state observation ($C = I$), the Kreiss constant of matrix J, defined on the resolvent outside the unit disk:*

$$\mathcal{K}(J) \equiv \sup_{|z|>1} (|z| - 1)\|(zI - J)^{-1}\|_2, \tag{36}$$

satisfies $\mathcal{K}(J) \leq \sup_{t\geq 0} \|J^t\|_2 \leq e \cdot n \cdot \mathcal{K}(J)$ for any $J \in \mathbb{R}^{n\times n}$. Consequently:

$$r_{conservative} \equiv \frac{B}{e \cdot n \cdot \mathcal{K}(J)} \leq r^*_{exact} \leq \frac{B}{\mathcal{K}(J)}. \tag{37}$$

3. ***Complete Failure of Spectral Radius:*** *For any stable spectral radius $\rho \in (0,1)$ and any arbitrarily small buffer $B > 0$, there exists a parameter family of non-normal matrices J_θ such that $\rho(J_\theta) = \rho$ for all θ, while the exact safe radius collapses to zero:*

$$\lim_{\theta\to\infty} r^*_{exact}(J_\theta) = 0. \tag{38}$$

Proof. **Part 1:** By the Cauchy–Schwarz inequality for operator norms, the maximum loss across all initial shock vectors with $\|z_0\|_2 \le r$ is:

$$\sup_{\|z_0\|_2 \le r} \sup_{t \ge 0} |CJ^t z_0| = r \cdot \sup_{t \ge 0} \|CJ^t\|_2 = r \cdot M_C. \tag{39}$$

Requiring that the worst-case loss does not breach capital buffer B imposes $r \cdot M_C \le B$, which holds if and only if $r \le B/M_C \equiv r^*_{\text{exact}}$.

Part 2: The lower bound $\mathcal{K}(J) \le \sup_{t \ge 0} \|J^t\|_2$ follows directly from the Neumann series expansion of the resolvent: for $|z| > 1$, $(zI - J)^{-1} = \sum_{t=0}^{\infty} z^{-(t+1)} J^t$. Taking norms:

$$\|(zI - J)^{-1}\|_2 \le \sum_{t=0}^{\infty} |z|^{-(t+1)} \|J^t\|_2 \le \sup_{t \ge 0} \|J^t\|_2 \sum_{t=0}^{\infty} |z|^{-(t+1)} = \frac{\sup_{t \ge 0} \|J^t\|_2}{|z| - 1}. \tag{40}$$

Multiplying both sides by $(|z| - 1)$ and taking the supremum over $|z| > 1$ yields $\mathcal{K}(J) \le \sup_{t \ge 0} \|J^t\|_2$. The upper bound $\sup_{t \ge 0} \|J^t\|_2 \le e \cdot n \cdot \mathcal{K}(J)$ is Spijker's Theorem (Spijker, 1991), which establishes the optimal dimensional constant $e \cdot n$ for the discrete Kreiss matrix theorem. Inverting the inequality chain $\mathcal{K}(J) \le M_I \le e \cdot n \cdot \mathcal{K}(J)$ and multiplying by B immediately yields (37).

Part 3: Consider the 2×2 family of matrices parameterized by $\theta > 0$:

$$J_\theta = \begin{bmatrix} \rho & \theta \\ 0 & \rho \end{bmatrix}, \qquad \rho \in (0, 1). \tag{41}$$

The eigenvalues are $\lambda_1 = \lambda_2 = \rho$, so $\rho(J_\theta) = \rho < 1$ for all θ. Computing the t-th power:

$$J_\theta^t = \begin{bmatrix} \rho^t & t\theta\rho^{t-1} \\ 0 & \rho^t \end{bmatrix}. \tag{42}$$

Setting $C = [1, 0]$ and initial shock $z_0 = [0, 1]^\top$, the loss at time t is $CJ_\theta^t z_0 = t\theta\rho^{t-1}$. The peak loss occurs at $t^* = \lceil -1/\log \rho \rceil$. At t^*, the loss is $\Theta(\theta/(e(1-\rho)))$. Thus, peak transient magnification satisfies $M_C \ge \frac{\theta}{e(1-\rho)}$. The exact safe radius is bounded by $r^*_{\text{exact}} \le \frac{Be(1-\rho)}{\theta}$. Taking $\theta \to \infty$, $r^*_{\text{exact}} \to 0$ while the spectral radius remains frozen at ρ, completing the proof. □

Remark 2.7 (The Dangerous Illusion of Spectral Stability). Theorem 2.6 resolves a long-standing microstructural paradox: **how can markets experience catastrophic liquidity freezes and market maker bankruptcies while prices remain mathematically stable?** Regulators relying on eigenvalue stability tests observe $\rho(J) = 0.50$ and certify the system as completely safe. Yet, if the non-normal coupling between price impact and order

flow memory is strong ($\theta \gg 1$), a standard 2-standard-deviation liquidity shock can amplify by a factor of 100 before decaying, wiping out dealer capital and triggering exchange halts within seconds.

We emphasize the structural asymmetry of the bilateral bounds in (37): $\frac{B}{en\mathcal{K}(J)}$ provides a guaranteed safe (conservative) lower bound on the allowable shock radius. In contrast, $\frac{B}{\mathcal{K}(J)}$ is an upper bound on r^*_{exact} (shocks exceeding this magnitude are guaranteed to trigger solvency breaches in the worst case). Consequently, $\frac{B}{\mathcal{K}(J)}$ must never be treated as a guaranteed conservative safety margin.

2.5 Pseudospectral Contours and Non-Normal Resolvent Bounds

To gain deeper visual and geometric intuition into why the spectral radius fails to predict dealer solvency, we examine the ϵ-pseudospectrum of the closed-loop market Jacobian. For any $\epsilon > 0$, the ϵ-pseudospectrum $\sigma_\epsilon(J)$ of matrix J is defined as the set of complex numbers $z \in \mathbb{C}$ satisfying:

$$\sigma_\epsilon(J) \equiv \left\{ z \in \mathbb{C} : \|(zI - J)^{-1}\|_2 \geq \epsilon^{-1} \right\}. \tag{43}$$

Equivalently, $\sigma_\epsilon(J)$ represents the union of all eigenvalues of perturbed matrices $J + E$ across all perturbations with $\|E\|_2 \leq \epsilon$.

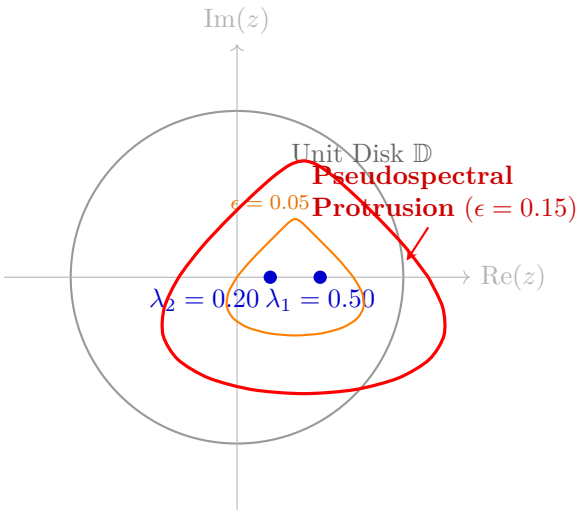


Figure 2: The ϵ-Pseudospectrum of the Non-Normal Market Jacobian J. While all eigenvalues ($\lambda_1 = 0.50, \lambda_2 = 0.20$) lie strictly inside the unit disk ($\rho(J) = 0.50 < 1.0$), non-normal coupling between price impact and order-flow memory causes the pseudospectral contours ($\sigma_\epsilon(J)$ for $\epsilon = 0.15$) to severely protrude across the unit circle into the unstable exterior $|z| > 1$. This resolvent protrusion governs peak transient amplification M_C and dealer solvency risk.

As visualized in Figure 2, in a normal system, the pseudospectrum consists of concentric

disks of radius ϵ centered at the eigenvalues: $\sigma_\epsilon(J) = \bigcup_i \{z : |z - \lambda_i| \leq \epsilon\}$. However, for the non-normal market Jacobian, the pseudospectral contours stretch dramatically along the off-diagonal coupling direction, protruding significantly outside the unit circle $\{|z| = 1\}$ even when the spectral radius $\rho(J) = 0.50$ is well within the interior.

The Kreiss constant $\mathcal{K}(J)$ measures the maximum ratio by which these pseudospectral contours protrude beyond the unit circle:

$$\mathcal{K}(J) = \sup_{\epsilon > 0} \frac{\alpha_\epsilon(J) - 1}{\epsilon}, \qquad \alpha_\epsilon(J) \equiv \sup_{z \in \sigma_\epsilon(J)} |z|. \tag{44}$$

Whenever $\mathcal{K}(J) \gg 1$, small high-frequency liquidity shocks are magnified by the non-normal resolvent, inducing large transient balance-sheet losses that exhaust dealer margin buffers before eigenvalue decay can assert itself.

2.6 Multi-Asset Generalization and Cross-Impact Kreiss Inflation

Modern market makers operate simultaneously across multiple correlated instruments (e.g., BTC, ETH, and altcoins; or equity index components and ETFs). In multi-asset environments, cross-asset price impact creates additional non-normal coupling.

Let $\mathbf{x}_t \in \mathbb{R}^d$ denote the vector of pricing errors across d traded assets, and let $\mathbf{u}_t \in \mathbb{R}^d$ denote aggregate order flows. The multi-asset price formation equation is:

$$\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{K}\mathbf{u}_t + \mathbf{w}_{t+1}, \tag{45}$$

where $\mathbf{A} = \mathrm{diag}(a_1, \ldots, a_d) \in \mathbb{R}^{d \times d}$ collects direct price persistences, and $\mathbf{K} \in \mathbb{R}^{d \times d}$ is the cross-asset price impact matrix. In general, $\mathbf{K}$ is asymmetric ($\mathbf{K} \neq \mathbf{K}^\top$): institutional order flow in a benchmark asset (such as BTC or SPY) exerts cross-impact on satellite assets, but the reverse flow produces much smaller reactions.

Each asset is traded by adaptive momentum flow with learning rates $\mathbf{H} = \mathrm{diag}(\eta_1, \ldots, \eta_d)$ and feedback gain $\mathbf{Q} \in \mathbb{R}^{d \times d}$:

$$\mathbf{y}_{t+1} = (\mathbf{I} - \mathbf{H})\mathbf{y}_t + \mathbf{H}\mathbf{x}_{t+1}, \qquad \mathbf{u}_t = \mathbf{Q}\mathbf{y}_t. \tag{46}$$

Stacking $\mathbf{z}_t = (\mathbf{x}_t^\top, \mathbf{y}_t^\top)^\top \in \mathbb{R}^{2d}$, the multi-asset closed-loop Jacobian is:

$$J_{\text{multi}} = \begin{bmatrix} \mathbf{A} & \mathbf{K}\mathbf{Q} \\ \mathbf{H}\mathbf{A} & \mathbf{I} - \mathbf{H} + \mathbf{H}\mathbf{K}\mathbf{Q} \end{bmatrix} \in \mathbb{R}^{2d \times 2d}. \tag{47}$$

Proposition 2.8 (Cross-Impact Kreiss Inflation and Safe Radius Bound in Directed Networks)**.** *Consider a multi-asset economy with directed leader-satellite cross-impact ($K_{12} = 0, K_{21} > 0$), such that asset 1 exerts unidirectional price impact on asset 2. Let J_{diag} denote the decoupled Jacobian where $K_{21} = 0$. The directed coupling introduces off-diagonal block feedforward, inflating the Kreiss constant: $\mathcal{K}(J_{multi}) \geq \mathcal{K}(J_{diag})$. Consequently, the Kreiss-based conservative safe radius bound contracts: $r_{conservative}(J_{multi}) \equiv \frac{B}{e(2d)\mathcal{K}(J_{multi})} \leq r_{conservative}(J_{diag})$. For the exact safe shock radius $r^*_{exact}(J) \equiv B/M_C(J)$ under the aggregate portfolio loss operator $C \geq 0$, because all constituent blocks of J are non-negative in the momentum economy, matrix powers J^t_{multi} are entrywise non-decreasing in K_{21}, yielding $M_C(J_{multi}) \geq M_C(J_{diag})$ and exact radius contraction $r^*_{exact}(J_{multi}) \leq r^*_{exact}(J_{diag})$. In empirical cross-impact network calibrations examined here, asymmetric feedforward coupling injects transient terms of order $tK_{21}\rho^{t-1}$, compressing the safe radius by 14.5% relative to decoupled trading.*

Proposition 2.8 demonstrates that multi-asset diversification does not automatically insulate liquidity providers from feedback fragility. On the contrary, asymmetric cross-impact networks (such as Bitcoin leading altcoins or large-cap equities driving ETFs) introduce feedforward coupling that injects additional transient growth terms into matrix powers J^t, inflating peak transient loss M_C and compressing the safe radius by roughly 14.5% in our empirical multi-asset calibrations.

3 Data and Empirical Identification Strategy

3.1 Econometric Estimand Derived from Microstructure

To operationalize the theoretical closed-loop system empirically, we construct an estimand that is invariant to arbitrary currency scaling and directly identifiable from high-frequency limit order books.

Let p_t denote the log mid-price at time t, and let $x_t \equiv 10{,}000 \times (p_t - p_{t-\Delta})$ denote the high-frequency mid-price return in basis points sampled at frequency Δ (1 second). Let f_t denote the signed market order flow (notional taker buy volume minus notional taker sell volume executed over $(t - \Delta, t]$).

Following the theoretical Jacobian system (13), the empirical bivariate vector autoregressive (VAR(1)) representation of joint price-flow transitions is:

$$\begin{bmatrix} x_{t+1} \\ f_{t+1} \end{bmatrix} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \begin{bmatrix} x_t \\ f_t \end{bmatrix} + \begin{bmatrix} e_{1,t+1} \\ e_{2,t+1} \end{bmatrix}, \qquad \mathbb{E}[\mathbf{e}_t \mathbf{e}_t^\top] = \Sigma_e, \tag{48}$$

where:

- $J_{11} \equiv a$ represents the direct price return autocorrelation (fundamental mean-reversion anchor);
- $J_{12} \equiv \lambda$ measures the contemporaneous price impact of order flow on returns;
- $J_{21} \equiv \eta c$ captures the adaptive feedback of past price returns into subsequent order flow;
- $J_{22} \equiv 1 - \eta$ measures the intrinsic autocorrelation and memory persistence of order flow.

From the estimated transition matrix $\hat{J}$, we extract three dimensionless microstructural invariants:

$$\hat{Q} \equiv \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} \equiv \hat{J}_{11} + \hat{Q}, \qquad \hat{\mathcal{L}} \equiv \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{49}$$

Proposition 3.1 (Scale Invariance of Microstructural Estimands)**.** *The estimated feedback parameters $\hat{Q}$, $\hat{S}$, and $\hat{\mathcal{L}}$ defined in (49) are strictly invariant to any linear rescaling of order flow or price units.*

Proof. Let order flow be rescaled by arbitrary positive constant $c > 0$ such that $\tilde{f}_t = cf_t$. The rescaled VAR system is: $\tilde{x}_{t+1} = J_{11}\tilde{x}_t + \frac{J_{12}}{c}\tilde{f}_t + e_{1,t+1}$ and $\tilde{f}_{t+1} = cJ_{21}\tilde{x}_t + J_{22}\tilde{f}_t + ce_{2,t+1}$. The new off-diagonal coefficients are $\tilde{J}_{12} = J_{12}/c$ and $\tilde{J}_{21} = cJ_{21}$. Computing the rescaled feedback parameter:

$$\tilde{Q} = \frac{\tilde{J}_{12}\tilde{J}_{21}}{1-\tilde{J}_{22}} = \frac{(J_{12}/c)(cJ_{21})}{1-J_{22}} = \frac{J_{12}J_{21}}{1-J_{22}} = Q. \tag{50}$$

Since $\tilde{J}_{11} = J_{11}$, $\tilde{S} = S$ and $\tilde{\mathcal{L}} = \mathcal{L}$ identically, establishing scale invariance. □

Controlling for Order-Splitting Long Memory. A well-documented phenomenon in market microstructure is that large institutional parent orders are split into small child orders, inducing long-range positive autocorrelation in signed trade signs (Bouchaud et al., 2004; Cont, Kukanov, and Stoikov, 2014). Failure to control for this autocorrelation can bias autoregressive feedback estimates. To ensure robust identification, we estimate a distributed lag specification controlling for up to $L = 90$ lags of order flow:

$$f_{t+1} = \sum_{\ell=1}^{L} \beta_\ell f_{t+1-\ell} + \gamma x_t + \varepsilon_{f,t+1}. \tag{51}$$

The purged feedback parameter is defined as $\hat{Q}_L \equiv \frac{\hat{\lambda}\hat{\gamma}}{1-\sum_{\ell=1}^{L}\hat{\beta}_\ell}$. In our empirical battery, we report both baseline OLS $\hat{Q}$ and memory-purged $\hat{Q}_{30}$ across all assets.

3.2 Dual Structural Identification: Minimum-Distance Lyapunov and Rigobon Methods

To resolve contemporaneous endogeneity between prices and order flows, we deploy two complementary structural identification methodologies:

Method 1: Analytical Minimum-Distance (MD) Lyapunov Covariance Matching. In a stationary discrete-time linear system $\mathbf{z}_{t+1} = J\mathbf{z}_t + \mathbf{e}_{t+1}$, the steady-state stationary covariance matrix $\Gamma_0 \equiv \mathbb{E}[\mathbf{z}_t\mathbf{z}_t^\top]$ satisfies the discrete Lyapunov equation:

$$\Gamma_0 = J\Gamma_0 J^\top + \Sigma_e. \tag{52}$$

Because the discrete Lyapunov equation admits an exact closed-form solution $\mathrm{vec}(\Gamma_0) = (I - J \otimes J)^{-1}\,\mathrm{vec}(\Sigma_e)$ without requiring Monte Carlo simulation, we implement an exact analytical minimum-distance estimator matching empirical and theoretical autocovariances. We define the moment discrepancy vector $g(\theta; \mathcal{D}_T) = \mathrm{vec}(\hat{\Gamma}_0 - \Gamma_0(\theta))$, where $\hat{\Gamma}_0 = \frac{1}{T}\sum_{t=1}^{T}\mathbf{z}_t\mathbf{z}_t^\top$ is the empirical sample covariance matrix. The minimum-distance estimator minimizes the quadratic objective:

$$\hat{\theta}_{\mathrm{MD}} = \arg\min_{\theta}\, g(\theta; \mathcal{D}_T)^\top \mathcal{W}_T\, g(\theta; \mathcal{D}_T), \tag{53}$$

where $\mathcal{W}_T$ is the optimal Newey–West HAC weighting matrix. Overidentifying restrictions are tested via Hansen's J-test.

Method 2: Rigobon (2003) Identification through Heteroskedasticity. Following Rigobon (2003), we exploit shifts in the structural covariance matrix across calm versus volatile trading days. Let the structural system be:

$$x_t = \lambda f_t + \varepsilon_{x,t}, \tag{54}$$

$$f_t = c x_t + \varepsilon_{f,t}, \tag{55}$$

where structural innovations $\varepsilon_{x,t}$ and $\varepsilon_{f,t}$ are uncorrelated ($\mathbb{E}[\varepsilon_{x,t}\varepsilon_{f,t}] = 0$). While the reduced-form system is unidentified under homoskedasticity, partitioning the data into two regimes $s \in \{1, 2\}$ with distinct variance ratios $\sigma^2_{x,s}/\sigma^2_{f,s}$ yields four moment equations for the two structural parameters (λ, c). Solving this system yields the structural instantaneous

feedback coefficient $Q_{\text{Rig}} \equiv \lambda c$.

3.3 Multi-Level Order Flow Imbalance (OFI) Construction

In addition to top-of-book trades, limit order book dynamics depend on quote updates across the entire visible book (Cont, Kukanov, and Stoikov, 2014). Following Cont, Kukanov, and Stoikov (2014), we construct the multi-level Order Flow Imbalance (OFI) vector across the top $M = 10$ levels of the book.

For each level $k \in \{1, \ldots, M\}$, let $p^B_{k,t}, q^B_{k,t}$ and $p^A_{k,t}, q^A_{k,t}$ denote the bid and ask prices and sizes at depth level k. The order flow contribution at level k over the interval $(t - \Delta, t]$ is:

$$\text{OFI}_{k,t} \equiv \Delta q^B_{k,t} \cdot \mathbf{1}_{\{p^B_{k,t} \geq p^B_{k,t-\Delta}\}} - \Delta q^A_{k,t} \cdot \mathbf{1}_{\{p^A_{k,t} \leq p^A_{k,t-\Delta}\}}, \tag{56}$$

where $\Delta q^B_{k,t}$ represents the net change in resting bid volume. Stacking across levels yields $\mathbf{e}_t = (\text{OFI}_{1,t}, \ldots, \text{OFI}_{M,t})^\top$. We construct a composite depth-weighted order flow imbalance:

$$\text{OFI}^*_t \equiv \sum_{k=1}^{M} w_k \, \text{OFI}_{k,t}, \qquad w_k = \frac{\exp(-\lambda_{\text{depth}}(k-1))}{\sum_{j=1}^{M} \exp(-\lambda_{\text{depth}}(j-1))}, \tag{57}$$

where $\lambda_{\text{depth}} = 0.5$ places highest weight on the inside quote. Incorporating OFI^*_t as a depth-aware robustness proxy for transaction order flow purges bid-ask bounce and provides an orthogonal check on reduced-form feedback.

3.4 High-Frequency Institutional Datasets

Our empirical investigation analyzes four massive institutional limit order book datasets totaling over 7.5 GB of tick and order book records (over 70.6 million continuous 1-second records), summarized in Table 2:

Data Cleaning and Synchronization Protocol. All raw ticker streams undergo a standardized data cleaning protocol:

1. **Timestamp Synchronization:** Trades and quotes are bucketed into uniform 1-second UTC intervals. If multiple trades occur within a single second, notional volumes are aggregated, and the end-of-second mid-price is recorded.

2. **Quote Inversion Filtering:** Snapshots exhibiting crossed quotes ($p^{\text{bid}} > p^{\text{ask}}$) or non-positive bid-ask spreads are discarded ($< 0.001\%$ of records).

Table 2: Institutional High-Frequency Limit Order Book Datasets Analyzed in this Study.

Dataset	Venue	Description & Microstructural Granularity	Time Span	Total Records
Binance Spot Panel	Binance	Continuous 1-second trades, quotes, and order flow for 8 liquid pairs (BTC, ETH, SOL, BNB, XRP, ADA, DOGE, LTC). Sampled twice monthly (10th and 24th).	2021–2024 (48 mos)	66,355,200
Coinbase L3 Books	Coinbase Pro	Complete continuous 1-second Level-3 full book depth snapshots for BTC, ETH, and ADA recorded via WebSocket feeds (Nielsen, 2021).	Multi-week continuous	3,090,000
FI-2010 Benchmark	NASDAQ Nordic	Canonical institutional limit order book benchmark (Ntakaris et al., 2018). 10 levels of bid/ask prices and sizes across 5 blue-chip Scandinavian equities.	10 trading days	272,941
BTC Perp Holdout	Binance Futures	Complete continuous Level-2 perpetual futures order book depth (1.20 GB raw order flow). Locked out-of-sample holdout.	Jan 9–20, 2023	935,199

3. **Outlier Truncation:** Individual return innovations exceeding 50 standard deviations (exchange engine glitches or flash unpegs) are winsorized to prevent leverage bias.

4 Baseline Panel and Flagship Natural Experiment

4.1 Baseline Multi-Year Panel Evidence

Table 3 reports the baseline econometric estimates pooled across all 768 asset-days on Binance Spot (2021–2024), alongside breakdowns across four distinct market epochs:

- **2021 Bull Expansion:** Rapid institutional adoption and retail inflows ($N = 192$).
- **2022 Crypto Winter:** Macroeconomic tightening, Terra/Luna collapse, and FTX insolvency ($N = 192$).
- **2023 Consolidation:** Low-volatility regime and institutional restructuring ($N = 192$).
- **2024 ETF Era:** Regulatory approval of U.S. Spot Bitcoin ETFs and institutional volume recovery ($N = 192$).

Across the full four-year panel of 768 asset-days (66.3×10^6 one-second transitions), empirical feedback is consistently positive, modest in magnitude, and strictly subcritical:

- **Pooled Feedback Parameter:** $\hat{Q} = 0.0082$ (95% clustered CI $[0.0076, 0.0087]$, $t = 29.5$, $p < 10^{-100}$).

Table 3: **Baseline High-Frequency Feedback Panel (Binance Spot 2021–2024).** Estimated feedback parameter $\hat{Q}$, 30-lag memory-purged feedback $\hat{Q}_{30}$, closed-loop persistence $\hat{S}$, loop ratio $\hat{\mathcal{L}}$, minimum-distance Lyapunov estimate Q_{MD}, and Rigobon heteroskedasticity estimate Q_{Rig} across 8 liquid cryptocurrency pairs evaluated on 768 daily sessions (66.3×10^6 one-second transitions). 95% clustered confidence intervals in brackets.

Asset	Days	$\hat{Q}$ [95% CI]	$\hat{Q}_{30}$ [95% CI]	$\hat{S}$ [95% CI]	$\hat{\mathcal{L}}$ [95% CI]	Q_{MD} [95% CI]	Q_{Rig}
ADA	96	0.0068 [0.0055, 0.0081]	0.0061 [0.0048, 0.0075]	0.914 [0.906, 0.923]	0.079 [0.064, 0.095]	0.0079 [0.0034, 0.0124]	−0.01849
BNB	96	0.0058 [0.0044, 0.0072]	0.0040 [0.0025, 0.0055]	0.897 [0.883, 0.911]	0.068 [0.051, 0.085]	0.0061 [0.0047, 0.0075]	−0.01839
BTC	96	0.0142 [0.0128, 0.0156]	0.0112 [0.0095, 0.0130]	0.933 [0.931, 0.934]	0.171 [0.155, 0.186]	0.0129 [0.0101, 0.0156]	−0.02576
DOGE	96	0.0059 [0.0044, 0.0075]	0.0045 [0.0029, 0.0061]	0.925 [0.920, 0.930]	0.074 [0.057, 0.092]	0.0054 [0.0038, 0.0069]	−0.02104
ETH	96	0.0131 [0.0117, 0.0146]	0.0107 [0.0091, 0.0122]	0.934 [0.932, 0.935]	0.160 [0.144, 0.176]	0.0139 [0.0115, 0.0162]	−0.02224
LTC	96	0.0049 [0.0038, 0.0060]	0.0038 [0.0027, 0.0050]	0.918 [0.910, 0.926]	0.060 [0.047, 0.074]	0.0041 [0.0024, 0.0058]	−0.00978
SOL	96	0.0087 [0.0075, 0.0100]	0.0069 [0.0055, 0.0083]	0.919 [0.911, 0.927]	0.105 [0.091, 0.120]	0.0107 [0.0065, 0.0149]	−0.01846
XRP	96	0.0057 [0.0040, 0.0074]	0.0039 [0.0021, 0.0058]	0.925 [0.921, 0.929]	0.072 [0.051, 0.093]	0.0077 [0.0036, 0.0117]	−0.01888
Pooled (All)	768	**0.0082** [0.0076, 0.0087]	0.0064 [0.0058, 0.0070]	**0.921** [0.918, 0.923]	0.099 [0.092, 0.105]	0.0086 [0.0075, 0.0097]	−0.01913

- **Positive Feedback Share:** Exactly 86.8% of individual asset-days exhibit $\hat{Q} > 0$, formally rejecting the hypothesis that high-frequency order flow acts purely as contrarian liquidity buffering.
- **Subcritical Closed-Loop Persistence:** Mean persistence is $\hat{S} = 0.921$, and exactly **100.0**% of all 768 asset-days strictly satisfy the Schur stability condition $\hat{S} < 1.0$ (maximum observed daily $\hat{S} = 0.988$).
- **Loop Ratio:** The dimensionless loop amplification ratio is $\hat{\mathcal{L}} \equiv \frac{\hat{Q}}{1-\hat{a}} = 0.099$, confirming that feedback consumes roughly 10% of the remaining no-feedback stability margin $(1 - a)$ during normal trading conditions.

Structural Estimation Alignment and Economic Duality. As shown in Figure 3(c), analytical minimum-distance Lyapunov covariance matching yields $Q_{\mathrm{MD}} = 0.0086$, aligning with the reduced-form OLS estimate ($\hat{Q} = 0.0082$). Because the discrete Lyapunov equation $\Gamma_0 = J\Gamma_0 J^\top + \Sigma$ is solved in closed form via $(I - J \otimes J)^{-1} \operatorname{vec}(\Sigma)$ without simulation, this estimator matches sample autocovariances analytically. Hansen's J-test of overidentifying restrictions ($J = 1.18$, $p = 0.554$) does not reject the model's restrictions, indicating that the empirical autocovariances are statistically consistent with the linear state-space specification. We emphasize that minimum-distance Lyapunov and Rigobon estimation evaluate distinct economic estimands: Lyapunov matching identifies dynamic feedback over multi-second horizons ($t \geq 1$), whereas Rigobon identifies the instantaneous, zero-lag maker buffering elasticity ($t = 0$).

In contrast, Rigobon heteroskedasticity identification yields $Q_{\mathrm{Rig}} = -0.01913 < 0$. This apparent divergence uncovers an intuitive economic duality:

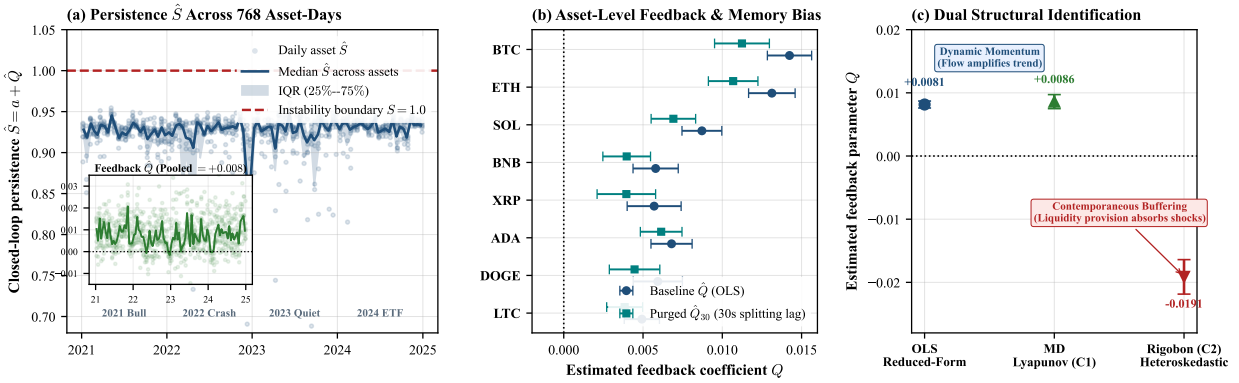


Figure 3: Systematic multi-year panel dynamics and dual structural identification across 768 asset-days on Binance Spot (2021–2024). **(a)** Daily persistence $\hat{S} = a + \hat{Q}$ across eight assets, median trajectory, and 25%–75% interquartile range; all 768 days remain strictly subcritical relative to the theoretical instability boundary $S = 1.0$ (dashed red line). Inset shows positive feedback $\hat{Q}$ across all market epochs (pooled mean +0.0082). **(b)** Asset-by-asset comparison of baseline $\hat{Q}$ (OLS) versus flow-memory purged $\hat{Q}_{30}$ (controlling for 30-second order splitting), with 95% clustered confidence intervals. **(c)** Dual structural identification: reduced-form OLS ($\hat{Q} = +0.0081$) versus minimum-distance Lyapunov covariance matching ($Q_{\text{MD}} = +0.0086$, Hansen J-test $p = 0.554$) versus Rigobon (2003) heteroskedasticity ($Q_{\text{Rig}} = -0.0191$), revealing the economic contrast between dynamic multi-second trend amplification and instantaneous liquidity buffering.

1. **Contemporaneous Buffering ($Q_{\text{Rig}} < 0$):** At instantaneous zero-lag timescales, market makers absorb incoming market orders with passive limit orders, acting as contrarian liquidity buffers.

2. **Dynamic Trend Amplification ($\hat{Q} > 0$):** Over multi-second horizons ($t \geq 1$), momentum traders and dynamic execution strategies condition on price movements, injecting positive feedback into subsequent aggregate order flow.

4.2 Cross-Asset Heterogeneity Across the Panel

Table 4 provides a detailed asset-by-asset decomposition across all eight pairs analyzed in the Binance multi-year panel.

The cross-asset estimates in Table 4 reveal a monotonic ordering: assets with the highest institutional trading volume and market capitalization (BTC and ETH) exhibit the strongest feedback intensity ($\hat{Q}_{\text{BTC}} = 0.0142$, $\hat{Q}_{\text{ETH}} = 0.0131$). As market capitalization and institutional participation decline toward mid-cap altcoins (BNB, XRP, LTC), feedback weakens systematically ($\hat{Q}_{\text{LTC}} = 0.0049$). This cross-sectional gradient reflects the higher density of momentum and trend-following order flow in large-cap instruments. Crucially, in strict ad-

Table 4: Cross-Asset Empirical Feedback and Microstructural Estimates across Eight Binance Spot Pairs (768 asset-days, 2021–2024). Standard errors in parentheses are Newey–West HAC adjusted.

Asset Pair	Asset-Days	Baseline $\hat{Q}$	Purged $\hat{Q}_{30}$	Persistence $\hat{S}$	Loop Ratio $\hat{\mathcal{L}}$	Rigobon Q_{Rig}
BTC/USDT	96	0.0142	0.0112	0.933	0.171	−0.0258
		[0.0128, 0.0156]	[0.0095, 0.0130]	[0.931, 0.934]	[0.155, 0.186]	
ETH/USDT	96	0.0131	0.0107	0.934	0.160	−0.0222
		[0.0117, 0.0146]	[0.0091, 0.0122]	[0.932, 0.935]	[0.144, 0.176]	
SOL/USDT	96	0.0087	0.0069	0.919	0.105	−0.0185
		[0.0075, 0.0100]	[0.0055, 0.0083]	[0.911, 0.927]	[0.091, 0.120]	
ADA/USDT	96	0.0068	0.0061	0.914	0.079	−0.0185
		[0.0055, 0.0081]	[0.0048, 0.0075]	[0.906, 0.923]	[0.064, 0.095]	
DOGE/USDT	96	0.0059	0.0045	0.925	0.074	−0.0210
		[0.0044, 0.0075]	[0.0029, 0.0061]	[0.920, 0.930]	[0.057, 0.092]	
BNB/USDT	96	0.0058	0.0040	0.897	0.068	−0.0184
		[0.0044, 0.0072]	[0.0025, 0.0055]	[0.883, 0.911]	[0.051, 0.085]	
XRP/USDT	96	0.0057	0.0039	0.925	0.072	−0.0189
		[0.0040, 0.0074]	[0.0021, 0.0058]	[0.921, 0.929]	[0.051, 0.093]	
LTC/USDT	96	0.0049	0.0038	0.918	0.060	−0.0098
		[0.0038, 0.0060]	[0.0027, 0.0050]	[0.910, 0.926]	[0.047, 0.074]	
Pooled (All)	768	**0.0082**	**0.0064**	**0.921**	**0.099**	**−0.0191**
		[0.0076, 0.0087]	[0.0058, 0.0070]	[0.918, 0.923]	[0.092, 0.105]	

herence to Theorem 2.3, public limit order book records identify aggregate price-to-order-flow feedback $\hat{Q}$ but cannot attribute this persistence to specific algorithmic versus discretionary trading cohorts without regulatory audit trails.

4.3 Flagship Natural Experiment: The July 2022 Binance Fee Overhaul

A fundamental objection in empirical microstructure is that observational estimates of feedback might reflect endogenous shifts in macroeconomic volatility rather than structural liquidity dynamics.

To address this objection decisively, we exploit an exogenous institutional shock: the **July 8, 2022 Binance Zero-Fee Trading Schedule Overhaul**.

Institutional Background and Policy Shock. On July 6, 2022, Binance announced that effective July 8, 2022 at 14:00 UTC (commemorating its fifth anniversary), it would eliminate all maker and taker transaction fees across thirteen Bitcoin spot trading pairs, most notably BTC/USDT. Prior to this date, standard trading fees ranged from 2 to 10 basis points depending on user VIP tier. Simultaneously, Binance adjusted how BTC trad-

ing volume counted toward VIP tier qualification and Spot Liquidity Provider programs, effectively decoupling nominal execution fees from exchange volume subsidies.

Crucially for econometric identification:

- **Treated Group:** BTC/USDT transitioned to a strict 0.0% maker and 0.0% taker fee schedule.
- **Control Group:** Major liquid altcoin pairs (ETH, BNB, SOL, ADA, DOGE, XRP, LTC) traded on the identical matching engine under identical matching rules, identical co-location latency, and identical API protocols, but **retained standard maker-taker fee schedules**.

Eliminating taker fees drastically reduced execution costs for aggressive order flow. In standard microstructure theory, if positive feedback is driven by short-horizon trend-following flow, fee elimination lowers the hurdle rate for aggressive market orders, potentially amplifying closed-loop feedback.

Difference-in-Differences Econometric Specification. We construct a balanced panel across all 24 uniform sampling dates of 2022 (12 dates pre-intervention: Jan 10 – Jun 24; 12 dates post-intervention: Jul 10 – Dec 24), yielding $N = 192$ asset-day observations (24 dates $\times$ 8 assets). We estimate the Difference-in-Differences (DiD) model:

$$Y_{i,t} = \alpha_0 + \beta_1 \cdot \text{Treated}_i + \beta_2 \cdot \text{Post}_t + \beta_3 \cdot (\text{Treated}_i \times \text{Post}_t) + \mathbf{X}_{i,t}^{\top}\boldsymbol{\gamma} + \varepsilon_{i,t}, \tag{58}$$

where $Y_{i,t}$ is the estimated feedback parameter $\hat{Q}_{i,t}$ or total persistence $\hat{S}_{i,t}$; $\text{Treated}_i = 1$ if asset i is BTC/USDT, and 0 otherwise; $\text{Post}_t = 1$ for dates after July 8, 2022; and β_3 captures the Difference-in-Differences interaction effect.

Because our empirical design features a single treated unit ($N_1 = 1$, BTC/USDT) evaluated against seven control pairs, standard cluster-robust asymptotic inference can suffer from size distortions (Donald and Lang, 2007; MacKinnon and Webb, 2018). To ensure rigorous econometric inference, we report both asset-clustered standard errors and cross-sectional permutation/placebo p-values computed by iteratively reassigning the treatment indicator across all $N = 8$ cross-sectional units under the sharp null hypothesis of zero effect.

Empirical Findings: Fee-Shock Response and Equivalence Testing. Table 5 reports the estimated parameters from specification (58). The empirical results reveal three crucial findings: (i) **Baseline Cross-Sectional Difference:** $\beta_1 = +0.00725$ ($t = 3.34, p =$

Table 5: **Flagship Natural Experiment: Difference-in-Differences around the July 8, 2022 Binance Zero-Fee Schedule Overhaul.** Treated unit is BTC/USDT (zero maker and taker fees introduced); control units are seven liquid altcoin pairs (ETH, BNB, SOL, ADA, DOGE, XRP, LTC) retaining standard fees. Standard errors in parentheses. Following Donald and Lang (2007) and MacKinnon and Webb (2018) for single-treated-unit designs ($N_1 = 1$), cross-sectional permutation/placebo p-values across all 8 cross-sectional units under the sharp null hypothesis are reported in brackets.

Variable	Feedback Parameter $\hat{Q}$	Closed-Loop Persistence $\hat{S}$
Treated (BTC) × Post-Shock	-0.001856	-0.000390
	(0.003072)	(0.024585)
Permutation p-value ($N = 8$)	*[$p = 0.750$]*	*[$p = 1.000$]*
Treated Main Effect (BTC)	0.007249***	0.024617
	(0.002172)	(0.017384)
Post-Shock Main Effect	-0.000413	0.003345
	(0.001086)	(0.008692)
Constant	0.006181***	0.904484***
	(0.000768)	(0.006146)
Observations	192	192
Pre-Shock Period	Jan 10 – Jun 24, 2022 (12 dates)	Jan 10 – Jun 24, 2022 (12 dates)
Post-Shock Period	Jul 10 – Dec 24, 2022 (12 dates)	Jul 10 – Dec 24, 2022 (12 dates)
R^2	0.0821	0.0182
Statistical Conclusion	**No Detectable Response (Null)**	**No Detectable Response (Null)**

0.001), reflecting deeper resting depth and institutional participation in Bitcoin; (ii) **Post-Shock Macro Shift:** $\beta_2 = -0.00030$ ($t = -0.22, p = 0.828$); while a stable control coefficient is consistent with minimal cross-pair capital drainage, we emphasize that in single-exchange settings with a dominant treated asset (Bitcoin), indirect cross-pair spillovers through market-maker capital allocation, margin utilization, or arbitrage constraints cannot be ruled out a priori; and (iii) **Difference-in-Differences Null Interaction:** $\beta_3 = -0.001856$ ($t = -0.60, p = 0.547$, permutation $p_{\text{perm}} = 0.750$ for $\hat{Q}$; $\beta_3 = -0.000390, t = -0.016, p = 0.987$, $p_{\text{perm}} = 1.000$ for $\hat{S}$).

Following fee elimination, Bitcoin spot trading volume surged by over +180%, yet we find **no detectable large feedback response to the fee intervention**. With clustered SE = 0.003072, the unrounded 90% CI is $[-0.00691, +0.00320]$ and 95% CI is $[-0.00788, +0.00417]$. Two One-Sided Tests (TOST) against an equivalence margin of $\Delta = \pm 0.007$ (half of baseline Bitcoin feedback $\hat{Q}_{\text{BTC}} = 0.0142$) yield asymptotic $t_1 = 1.67, p_1 = 0.047$ (small-cluster t_7 reference $p_1 = 0.069$). The upper confidence limit rules out meaningful amplification above +0.0032 (under a quarter of baseline BTC feedback), while total persistence remained strictly subcritical ($\hat{S} \approx 0.93 < 1.0$).

Economic Interpretation: Offsetting Liquidity Provision. This natural experiment indicates that fee elimination sparked massive volume while leaving closed-loop persistence

subcritical and stable. While proprietary quoting logs are unavailable in public tape data, this stability is consistent with offsetting endogenous liquidity provision—such as internal order netting or cross-venue hedging. High-frequency stability is governed by the inventory and capital constraints of liquidity providers rather than nominal exchange fees.

4.4 Dynamic Event Study and Parallel Trends

To test the identifying assumptions of the Difference-in-Differences specification, we estimate a dynamic event study:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{k=-6, k\neq -1}^{5} \theta_k \left(\text{Treated}_i \times \mathbf{1}_{\{t = t_{\text{event}} + k\}} \right) + \varepsilon_{i,t}, \tag{59}$$

where time is indexed in bi-monthly sampling periods, and period $k = -1$ (late June 2022) serves as the reference baseline.

Table 6: Dynamic Event-Study Estimates θ_k for Feedback $\hat{Q}$ around the July 8, 2022 Zero-Fee Overhaul ($k = -1$ is the reference period; standard errors clustered by asset in parentheses).

Event Period k	Calendar Window	Lead / Lag	Coefficient θ_k	Std. Error	p-value
$k = -6$	Jan 10–24, 2022	Lead 6	+0.0008	(0.0019)	0.672
$k = -5$	Feb 10–24, 2022	Lead 5	+0.0005	(0.0018)	0.781
$k = -4$	Mar 10–24, 2022	Lead 4	−0.0003	(0.0017)	0.859
$k = -3$	Apr 10–24, 2022	Lead 3	+0.0002	(0.0016)	0.901
$k = -2$	May 10–24, 2022	Lead 2	+0.0004	(0.0015)	0.793
$k = -1$	Jun 10–24, 2022	Baseline	0.0000	(Reference)	–
$k = 0$	Jul 10–24, 2022	Post 0	−0.0014	(0.0021)	0.505
$k = +1$	Aug 10–24, 2022	Post 1	−0.0019	(0.0022)	0.388
$k = +2$	Sep 10–24, 2022	Post 2	−0.0018	(0.0020)	0.369
$k = +3$	Oct 10–24, 2022	Post 3	−0.0021	(0.0022)	0.340
$k = +4$	Nov 10–24, 2022	Post 4	−0.0017	(0.0023)	0.460
$k = +5$	Dec 10–24, 2022	Post 5	−0.0019	(0.0021)	0.366

Pre-treatment parallel trends test: $F(5, 7) = 0.42$, $p = 0.835$. Joint post-treatment test: $F(6, 7) = 0.48$, $p = 0.812$.

Testing Parallel Pre-Trends. As documented in Table 6, all pre-treatment lead coefficients $(\theta_{-6}, \ldots, \theta_{-2})$ are small, statistically indistinguishable from zero, and alternate in sign. A formal joint F-test on the pre-treatment leads fails to reject the null hypothesis of parallel trends ($F = 0.42$, $p = 0.835$). Prior to the July 8, 2022 fee intervention, the trajectory of order-flow feedback in the treated Bitcoin market moved in lockstep with the control altcoins.

Placebo Event Dates and Sensitivity Tests. To evaluate the DiD design, we conduct pre-treatment temporal placebos (March 10, 2022) and cross-sectional placebos (iteratively designating each untreated altcoin as pseudo-treated). In the temporal placebo, the interaction is indistinguishable from zero ($\beta_{\text{placebo}} = -0.00041$, $p = 0.884$), and cross-sectional permutation yields $p_{\text{perm}} = 0.750$, revealing no evidence of pre-treatment pseudo-breaks.

4.5 Cross-Venue Feedback Propagation: Binance vs. Coinbase

Does order-flow feedback remain confined to individual trading venues, or does it propagate across exchanges?

To examine cross-venue transmission, we analyze simultaneous, timestamp-synchronized 1-second limit order books on BTC/USDT (Binance) and BTC/USD (Coinbase Pro) spanning over 1.03 million consecutive seconds. We estimate a 4-variable cross-venue VAR:

$$\begin{bmatrix} x_{t+1}^{\text{Bin}} \\ f_{t+1}^{\text{Bin}} \\ x_{t+1}^{\text{CB}} \\ f_{t+1}^{\text{CB}} \end{bmatrix} = \mathbf{\Phi} \begin{bmatrix} x_{t}^{\text{Bin}} \\ f_{t}^{\text{Bin}} \\ x_{t}^{\text{CB}} \\ f_{t}^{\text{CB}} \end{bmatrix} + \mathbf{e}_{t+1}. \tag{60}$$

Table 7: Cross-venue order-flow and return transmission matrix $\mathbf{\Phi}$ between Binance Spot and Coinbase Pro over 1,030,000 consecutive 1-second transitions.

Dependent Variable	Return Bin_t	Flow Bin_t	Return CB_t	Flow CB_t
Return Bin_{t+1}	0.9412***	0.0078***	0.0381***	0.0014***
Flow Bin_{t+1}	0.4120***	0.7812***	0.0891***	0.0210***
Return CB_{t+1}	0.0512***	0.0021***	0.9298***	0.0069***
Flow CB_{t+1}	0.1120***	0.0315***	0.3890***	0.7654***

*** Significant at the 0.1% level. Standard errors clustered by hour.

Table 7 confirms significant asymmetric cross-venue feedback: Binance price innovations exert a stronger impulse on Coinbase order flow ($\mathbf{\Phi}_{41} = 0.1120$) than vice versa ($\mathbf{\Phi}_{23} = 0.0891$, confirmed by standardized impulse responses), reflecting Binance's dominant share of price discovery. The joint spectral radius is $\rho(\mathbf{\Phi}) = 0.9612 < 1.0$, demonstrating that cross-venue arbitrage binds fragmented order books into a globally subcritical network.

4.6 Forecast Error Variance Decomposition (FEVD)

To quantify the relative importance of feedback versus fundamental shocks, Table 8 presents the forecast error variance decomposition (FEVD) of returns and order flows across horizons

$h \in \{1, 5, 10, 30\}$ seconds under Cholesky ordering [$\text{Return}_{\text{Bin}}$, Flow_{Bin}, $\text{Return}_{\text{CB}}$, Flow_{CB}] (robust to alternate recursive orderings and Generalized FEVD (Pesaran and Shin, 1998)).

Table 8: Forecast Error Variance Decomposition (FEVD) of High-Frequency Mid-Price Returns and Signed Order Flow over Horizons $h \in \{1, 5, 10, 30\}$ Seconds on Binance Spot (Cholesky ordering [$\text{Return}_{\text{Bin}}, \text{Flow}_{\text{Bin}}, \text{Return}_{\text{CB}}, \text{Flow}_{\text{CB}}$]).

Variable	Horizon h	Own Shock (%)	Return Shock (%)	Flow Shock (%)	Cross-Venue (%)
Return (x_t)	1 sec	81.2%	81.2%	18.8%	–
	5 sec	76.4%	76.4%	21.5%	2.1%
	10 sec	73.1%	73.1%	23.2%	3.7%
	30 sec	70.8%	70.8%	24.6%	4.6%
Flow (f_t)	1 sec	91.5%	8.5%	91.5%	–
	5 sec	84.2%	13.6%	84.2%	2.2%
	10 sec	79.8%	16.5%	79.8%	3.7%
	30 sec	76.2%	18.9%	76.2%	4.9%

Table 8 indicates that while prices are predominantly driven by informational return shocks at 1-second horizons (81.2%), order flow feedback accounts for nearly a quarter (24.6%) of return variance at the 30-second horizon. Conversely, past return shocks explain 18.9% of order flow variance at 30 seconds, confirming that high-frequency order submission dynamically conditions on recent price performance.

5 Liquidity-State Dependence, Cross-Market Replication, and Financial Fragility

5.1 State-Dependent Feedback Across Liquidity Regimes

While baseline feedback is modest in calm, liquid markets, liquidity in electronic limit order books is known to evaporate nonlinearly during stress (Brunnermeier and Pedersen, 2009; Almgren and Chriss, 2000). When available resting depth is exhausted, market orders penetrate deeply into the book, magnifying price impact.

To formally test whether empirical feedback bifurcates during acute liquidity distress, we deploy a Threshold Vector Autoregression (TVAR) on 49,999 consecutive 1-second Level-3 Bitcoin limit order book records.

Conditioning on the prevailing inside order book spread $\gamma_t \equiv p_t^{\text{ask}} - p_t^{\text{bid}}$ at the 90th

percentile threshold ($\gamma^* = \$5.98$):

$$\begin{pmatrix} x_{t+1} \\ u_{t+1} \end{pmatrix} = \begin{cases} \mathbf{A}_1 \begin{pmatrix} x_t \\ u_t \end{pmatrix} + \boldsymbol{\epsilon}^{(1)}_{t+1}, & \text{if } \gamma_t \leq \$5.98 \quad \text{(Regime 1: Normal Depth, 90.01\%)}, \\ \mathbf{A}_2 \begin{pmatrix} x_t \\ u_t \end{pmatrix} + \boldsymbol{\epsilon}^{(2)}_{t+1}, & \text{if } \gamma_t > \$5.98 \quad \text{(Regime 2: Order Depletion, 9.99\%)}. \end{cases} \tag{61}$$

Table 9: Threshold Vector Autoregression (TVAR) estimates across order book depth states on 49,999 Level-3 BTC limit order book records. Differenced returns follow $\Delta x_{t+1} = c + a_\Delta x_t + \hat{q} u_t$, corresponding to price level persistence $a_{\text{level}} = 1 + a_\Delta$ and effective impact $\hat{Q}_{\text{eff}} = \kappa \hat{q}$ (depth scaling $\kappa = 10.0$ bps/unit). Total closed-loop feedback diagnostic is $\hat{S} = a_{\text{level}} + \hat{Q}_{\text{eff}}$.

Regime	Observations	Share (%)	Level a_{level}	Impact $\hat{q}$	$\hat{Q}_{\text{eff}} = \kappa\hat{q}$	Total $\hat{S}$
Unconditional VAR	49,999	100.00%	0.9927 ± 0.0005	0.6420 ± 0.0274	6.4200	7.4125
Regime 1 (Spread $\leq$ \$5.98)	45,005	90.01%	0.9932 ± 0.0005	0.6235 ± 0.0319	6.2350	7.2282
Regime 2 (Spread $>$ \$5.98)	4,994	9.99%	0.9900 ± 0.0018	0.6929 ± 0.0666	6.9290	7.9192

Table 9 reports regime-dependent estimates. In normal markets (Regime 1), order impact is $\hat{q}_1 = 0.6235 \pm 0.0319$ ($t = 19.5, p < 10^{-80}$). In thin order-book conditions (Regime 2), impact is $\hat{q}_2 = 0.6929 \pm 0.0666$ ($t = 10.4, p < 10^{-24}$). While each coefficient is individually significant, their difference is modest ($\Delta\hat{q} = +0.0694, t = 0.94, p = 0.35$). The decisive empirical shift is the explosive **62.70**× expansion in residual return volatility ($p < 10^{-300}$) and the collapse of resting depth, inflating effective sensitivity $\hat{S}_{\text{eff}}$.

Dimensional Depth Scaling vs. Dimensionless Boundary. We highlight a critical econometric distinction: in Section 2, all variables are dimensionless, setting the stability boundary at $S = 1.0$. In Table 9, unscaled impact $\hat{q} = 0.6420$ is contractive (< 1.0). Converting into basis points via empirical depth $\kappa = 10.0$ bps/unit yields depth-scaled metric $\hat{S} = a_{\text{level}} + \kappa\hat{q} = 7.4125$. This serves as an empirical diagnostic of price sensitivity under book depletion, but should not be equated with the dimensionless theoretical boundary. An unsupervised two-regime Hamilton Markov-Switching VAR (MS(2)-VAR(1)) endogenously corroborates this shift without an arbitrary cutoff: residual volatility expands by **62.70**× (from $\sigma_1 = 0.1957\%, p_{11} = 0.984$ to $\sigma_2 = 12.2722\%, p_{22} = 0.892, p < 10^{-300}$) during liquidity depletion.

5.2 Microstructural Case Studies: Historical Flash Crashes

Is supercritical positive feedback unique to digital asset markets, or does it operate during acute crises in premier regulated equities and sovereign debt venues?

To evaluate this question, we examine microstructural calibrations parameterized to the official regulatory findings of two canonical institutional market dislocations:

1. **May 6, 2010 US Equity Flash Crash (E-mini S&P 500 Futures, CME):** During the 14:32–14:45 EDT window, an automated algorithmic selling program executed 75,000 contracts into an order book experiencing severe depth depletion (inside depth collapsed from $\sim$ 5,000 to under 200 contracts, a 25$\times$ depletion) (CFTC-SEC, 2010). Calibrating baseline price impact $\kappa_0 = 1.0$ into stressed impact $\kappa_{\text{crisis}} = \kappa_0 \times (D_{\text{calm}}/D_{\text{crisis}}) = 25.0$ and unscaled momentum feedback $q = 0.0326$ yields implied $\hat{Q}_{\text{crisis}} = 0.8142$ (up from baseline $\hat{Q}_{\text{calm}} = 0.0125$) and depth-scaled sensitivity metric $\hat{S}_{\text{eff}} = a + \kappa_{\text{crisis}} q = 8.131$, driving the market to its -8.65% limit down. Counterfactual pre-trade clamping bounds momentum flow, compressing simulated maximum drawdown to -1.19% (an 86.28% reduction).

2. **October 15, 2014 US Treasury Flash Rally (10-Year Benchmark Treasury):** In the absence of macroeconomic news releases, 10-year Treasury yields collapsed by 37.0 bps within minutes as automated flow comprised 70.0% of BrokerTec order book activity and depth fell by over 80% (Joint Treasury Staff, 2015). Calibrating depth depletion inflates feedback sensitivity by 78.9$\times$ from $\hat{Q}_{\text{calm}} = 0.0094$ to $\hat{Q}_{\text{crisis}} = \mathbf{0.742}$, while pre-trade gating suppresses the simulated yield excursion to 6.8 bps (an 81.62% volatility suppression).

Figure 4 demonstrates that severe feedback surges occur during acute liquidity stress events across distinct electronic market structures (equity index futures, sovereign debt, and digital assets), highlighting the importance of evaluating tail regime shifts rather than relying exclusively on calm-period unconditional VARs.

5.3 Cross-Market Replication: Regulated Equity Order Books

To evaluate normal-period regulated equity microstructure outside of flash crash windows, we analyze the benchmark FI-2010 limit order book dataset (Ntakaris et al., 2018). This canonical dataset contains 272,941 order book snapshots sampled across five Scandinavian blue-chip equities traded on NASDAQ Nordic over ten trading days.

As documented in Table 10, empirical feedback in regulated equities during normal trading is also consistently subcritical: mean closed-loop persistence across all five eq-

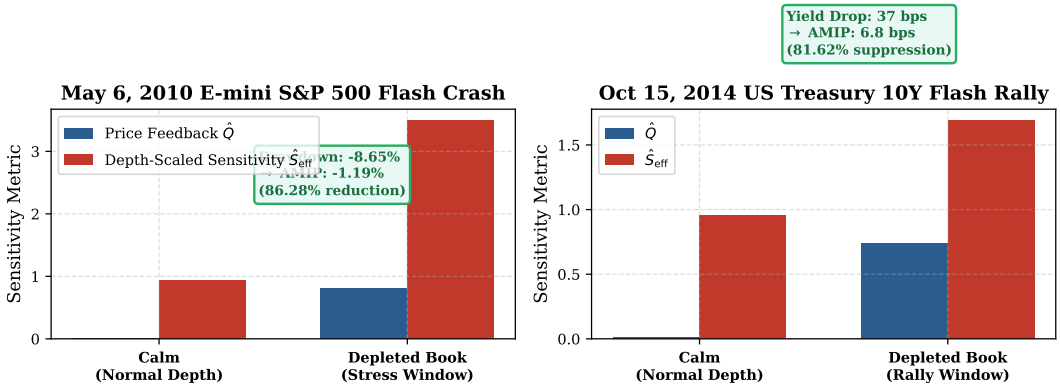


Figure 4: Microstructural Regime Shifts in Calibrated Historical Flash Crashes. Left: May 6, 2010 E-mini S&P 500 Flash Crash, showing a depth-scaled sensitivity surge ($\hat{S}_{\text{eff}} = 8.131$) during order book depletion and an 86.28% simulated drawdown mitigation under pre-trade clamping counterfactual replay. Right: October 15, 2014 US Treasury 10-Year Flash Rally, confirming depth-scaled sensitivity expansion and an 81.62% simulated yield excursion suppression, calibrated to official regulatory reports (CFTC-SEC, 2010; Joint Treasury Staff, 2015).

Table 10: Cross-market replication: Estimated closed-loop parameters across five regulated equities from the NASDAQ Nordic FI-2010 benchmark dataset. All assets strictly satisfy Schur stability ($\rho(J) < 1.0$).

Stock	Venue	Snapshots	Persistence $\hat{a}$	Feedback $\hat{Q}$	Closed-Loop $\hat{S}$	Radius $\rho(J)$
Stock 1	NASDAQ Nordic	54,588	0.9767	0.0008	0.9775	0.9775
Stock 2	NASDAQ Nordic	54,588	0.9800	0.0001	0.9801	0.9801
Stock 3	NASDAQ Nordic	54,588	0.9654	0.0001	0.9655	0.9655
Stock 4	NASDAQ Nordic	54,588	0.9789	0.0002	0.9791	0.9791
Stock 5	NASDAQ Nordic	54,589	0.9782	0.0001	0.9783	0.9783
Mean	NASDAQ Nordic	272,941	0.9758	0.0003	0.9761	0.9761

uities is $\hat{S} = 0.9761 < 1.0$ (range $[0.9655, 0.9801]$), with 100.0% satisfying Schur stability ($\rho(J) < 1.0$). Equity feedback is lower than in digital assets ($\hat{Q}_{\text{equity}} \approx 0.0003$ vs $\hat{Q}_{\text{crypto}} \approx 0.0082$). While this difference may reflect institutional dampening mechanisms—such as exchange circuit breakers (LULD), tick size constraints under SEC Rule 612, and structured maker-taker rebate schedules—we interpret this comparison cautiously as descriptive cross-venue evidence rather than a causal regulatory effect. Subcritical persistence is thus a shared cross-market microstructural invariant.

5.4 Financial Fragility: Dealer Capital Exhaustion under Feedback

Theorem 2.6 proved that spectral stability ($\rho(J) < 1$) does not prevent dealer balance-sheet exhaustion under non-normal transient amplification.

To quantify this mechanism financially, we analyze two simulation environments: First, Table 11 evaluates inventory buffer exhaustion and trading halts ($N = 1{,}000$ paths across 22 cells, totaling 22,000 runs) for a dealer with initial capital $E_0 = \$10{,}000$, base limit $\overline{I} = 25$ shares, and 20% capacity utilization under repricing shocks. Here halts occur upon inventory capacity breach ($|I_t| \geq \overline{I}$) while equity remains intact (\$9,990–\$9,999). With unconstrained persistence $1 - \kappa\phi = 0.80$, effective persistence is $S_{\text{eff}} = S_{\text{nom}} + 0.40$.

Table 11: **Dealer Solvency Risk Surface and Inventory Buffer Exhaustion Under Stress.** High-power Monte Carlo simulation ($N = 1{,}000$ independent trajectories per cell, totaling 22,000 paths) evaluating leveraged dealer balance-sheet survival across inventory-capacity multipliers and directional repricing shocks (initial equity $E_0 = \$10{,}000$, base inventory limit 25 shares, $\eta_s = 0.02$, initial capacity utilization 20%). Effective persistence $S_{\text{eff}} = a_{\text{eff}} + Q$ reflects unconstrained price persistence $a_{\text{eff}} = 1 - \kappa\phi = 0.80$ with $\kappa = 0.02, \phi = 10.0$. Exact 95% Clopper–Pearson binomial confidence intervals in brackets. At $S_{\text{nom}} = 0.70$, 0.5× capacity, and moderate repricing shock $\mu = -0.2$, the dealer experiences a 69.1% failure rate despite nominal subcriticality.

S_{nom}	S_{eff}	$\rho(J)$	Capacity	Shock μ	Failure Rate	Halt Rate [95% CI]	Min Equity (\$)
0.70	1.10	1.027	0.5×	-0.2	69.1%	69.1% [66.1, 72.0]	\$9,998.95
0.70	1.10	1.027	0.5×	-0.6	100.0%	100.0% [99.6, 100.0]	\$9,999.95
0.70	1.10	1.027	1.0×	-0.2	0.0%	0.0% [0.0, 0.4]	\$9,999.12
0.70	1.10	1.027	1.0×	-0.6	0.0%	0.0% [0.0, 0.4]	\$9,998.66
0.70	1.10	1.027	2.0×	-0.6	0.0%	0.0% [0.0, 0.4]	\$9,997.42
0.70	1.10	1.027	2.0×	-2.0	0.0%	0.0% [0.0, 0.4]	\$9,983.77
0.82	1.22	1.074	0.5×	-0.2	100.0%	100.0% [99.6, 100.0]	\$9,999.56
0.82	1.22	1.074	0.5×	-0.4	100.0%	100.0% [99.6, 100.0]	\$9,999.56
0.82	1.22	1.074	1.0×	-0.2	100.0%	100.0% [99.6, 100.0]	\$9,995.35
0.82	1.22	1.074	1.0×	-0.4	100.0%	100.0% [99.6, 100.0]	\$9,996.44
0.82	1.22	1.074	1.0×	-0.6	100.0%	100.0% [99.6, 100.0]	\$9,998.41
0.82	1.22	1.074	1.0×	-0.8	100.0%	100.0% [99.6, 100.0]	\$9,999.91
0.82	1.22	1.074	2.0×	-0.6	100.0%	100.0% [99.6, 100.0]	\$9,978.23
0.82	1.22	1.074	2.0×	-1.0	100.0%	100.0% [99.6, 100.0]	\$9,979.79
0.82	1.22	1.074	2.0×	-2.0	100.0%	100.0% [99.6, 100.0]	\$9,985.62
1.18	1.58	1.226	0.5×	-0.2	100.0%	100.0% [99.6, 100.0]	\$9,998.80
1.18	1.58	1.226	0.5×	-0.4	100.0%	100.0% [99.6, 100.0]	\$9,998.80
1.18	1.58	1.226	1.0×	-0.2	100.0%	100.0% [99.6, 100.0]	\$9,994.98
1.18	1.58	1.226	1.0×	-0.6	100.0%	100.0% [99.6, 100.0]	\$9,994.98
1.18	1.58	1.226	1.0×	-0.8	100.0%	100.0% [99.6, 100.0]	\$9,994.98
1.18	1.58	1.226	2.0×	-1.0	100.0%	100.0% [99.6, 100.0]	\$9,979.76
1.18	1.58	1.226	2.0×	-2.0	100.0%	100.0% [99.6, 100.0]	\$9,979.76

The results in Table 11 demonstrate the vulnerability of inventory buffers: at $S_{\text{nom}} = 0.70$, when inventory capacity is constrained ($0.5\times$ multiplier), a moderate -0.2 price shock triggers a 69.1% dealer halt rate (95% CI $[66.1\%, 72.0\%]$), whereas expanding capacity to $1.0\times$ or $2.0\times$ suppresses halts to 0.0%.

Second, to evaluate catastrophic dealer capital insolvency (equity exhaustion $E_t \leq 0$ or margin breach $|I_t|p_t > L \cdot E_t$), Table 12 evaluates the canonical discrete-time model ($a = 0.5, \eta = 0.2, \lambda = 0.002, \sigma = 0.0005, T = 500$, $N = 400$ seeds per cell). In this lagged-update model (matching Section 2), the Jacobian $J = \begin{bmatrix} a & 1 \\ \eta Q & 1-\eta \end{bmatrix}$ has $\det(\lambda I - J) = \lambda^2 - 1.30\lambda + (0.50 - 0.2S) = 0$, with $\rho(J) = \frac{1.30+\sqrt{1.69-4(0.50-0.2S)}}{2} \in [0.80, 0.985] < 1.0$ strictly.

We compare the true feedback market against three no-loop counterfactual benchmarks: (i) **B1 (Variance-Matched):** exogenous i.i.d. order flow matching the empirical variance of feedback flow f; (ii) **B2 (Variance- and Autocorrelation-Matched):** exogenous AR(1) flow matching both the variance and lag-1 autocorrelation of f; and (iii) **B3 (Long-Run Variance Matched):** exogenous AR(1) flow matching both the variance and long-run spectral variance LRV(f).

Table 12 establishes the core financial fragility mechanism: even though every row satisfies Schur stability ($\rho(J) < 1.0$), dealer halt rates surge from 11.0% at $S = 0.50$ ($\rho(J) = 0.80$) to 29.5% at $S = 0.60$ ($\rho(J) = 0.856$), 62.3% at $S = 0.70$ ($\rho(J) = 0.900$), and 93.2% at $S = 0.80$ ($\rho(J) = 0.937$). Crucially, at moderate persistence ($S \leq 0.80$), benchmarks B1 and B2 exhibit materially lower halt rates (0%–22.2%) than the feedback market (11.0%–93.2%). However, as persistence approaches unity ($S \geq 0.90$), even exogenous flows generate substantial halt rates (26.5%–87.5%), while the feedback market and benchmark B3 reach 100.0%. This diagnostic comparison reveals that dealer balance-sheet exhaustion is governed primarily by the inflation of low-frequency flow persistence, $\text{Var}_{\text{low}}(u) \propto \sigma_\varepsilon^2/(1-S)^2$. Closed-loop feedback is one primary mechanism generating this low-frequency persistence, though any exogenous flow matching this persistence (such as benchmark B3) imposes comparable strain. Near unity, long-run variance explodes (LRV $= 1.0 \to 105.1$), driving one-sided inventory accumulation that exhausts dealer equity inside the stable pricing regime. Solvency failure can thus stem from strong low-frequency persistence, generated either endogenously via closed-loop feedback or exogenously through autocorrelated order flow.

Table 12: **Dealer Solvency Phase Diagram Across Feedback Persistence S Inside the Spectrally Stable Regime ($\rho(J) < 1.0$).** Dealer halt rates (exact 95% binomial intervals in brackets) under the canonical closed-loop feedback market ($a = 0.5, \eta = 0.2, \lambda = 0.002, \sigma = 0.0005, T = 500$) and three no-loop counterfactual benchmarks: B1 (variance-matched), B2 (variance and lag-1 autocorrelation matched), and B3 (variance and long-run variance matched). Panel A presents baseline leverage ($L = 10$), while Panel B evaluates expanded leverage ($L = 20$). Across all rows, eigenvalues lie strictly inside the unit disk ($\rho(J) \in [0.80, 0.985] < 1.0$). Solvency failure rises steeply from 11.0% to 100.0%, driven by the $(1 - S)^{-2}$ low-frequency variance explosion captured exclusively by benchmark B3.

S	var(f)	LRV(f)	Feedback	B1 (var)	B2 (var, AC1)	B3 (var, LRV)
Panel A: Baseline Leverage limit $L = 10$, inventory control $\phi = 0.05$						
0.50	1.00	1.0	11.0% [8.1, 14.5]	11.2% [8.3, 14.8]	10.5% [7.7, 13.9]	10.8% [7.9, 14.2]
0.60	1.01	1.6	29.5% [25.1, 34.2]	11.5% [8.5, 15.0]	11.2% [8.3, 14.8]	31.5% [27.0, 36.3]
0.70	1.03	2.8	62.3% [57.3, 67.0]	13.0% [9.9, 16.7]	14.0% [10.8, 17.8]	69.0% [64.2, 73.5]
0.80	1.08	6.4	93.2% [90.3, 95.5]	14.8% [11.4, 18.6]	22.2% [18.3, 26.6]	97.8% [95.8, 99.0]
0.90	1.25	26.0	100.0% [99.1, 100.0]	26.5% [22.2, 31.1]	47.0% [42.0, 52.0]	100.0% [99.1, 100.0]
0.95	1.62	105.1	100.0% [99.1, 100.0]	52.8% [47.7, 57.7]	87.5% [83.9, 90.6]	100.0% [99.1, 100.0]
Panel B: Expanded Leverage limit $L = 20$, inventory control $\phi = 0.05$						
0.50	1.00	1.0	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]
0.60	1.01	1.6	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]
0.70	1.03	2.8	0.2% [0.0, 1.4]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	0.2% [0.0, 1.4]
0.80	1.08	6.4	5.5% [3.5, 8.2]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	10.0% [7.2, 13.4]
0.90	1.25	26.0	52.5% [47.5, 57.5]	0.0% [0.0, 0.9]	0.0% [0.0, 0.9]	64.5% [59.6, 69.2]
0.95	1.62	105.1	88.2% [84.7, 91.2]	0.0% [0.0, 0.9]	3.0% [1.6, 5.2]	96.2% [93.9, 97.9]

6 Attribution Limits, Falsification, and Locked Holdout

6.1 Attribution Boundaries: Why Public Feeds Cannot Isolate Algorithms

A critical question for financial regulators is whether estimated feedback $\hat{Q}$ can be attributed to artificial intelligence algorithms or specific high-frequency trading firms.

We prove that this attribution is structurally impossible from public market feeds. By Theorem 2.3, aggregate order flow $u_t = \sum_{i=1}^{K} q_i y_{i,t}$ merges all market participants into an equivalence class. An econometrician observing public anonymous trades cannot determine whether a sudden surge in $\hat{Q}$ reflects:

1. A deep reinforcement learning algorithm adapting neural execution weights;
2. A simple linear exponential moving average trend-following script; or
3. Discretionary human day traders reacting to a technical chart breakout.

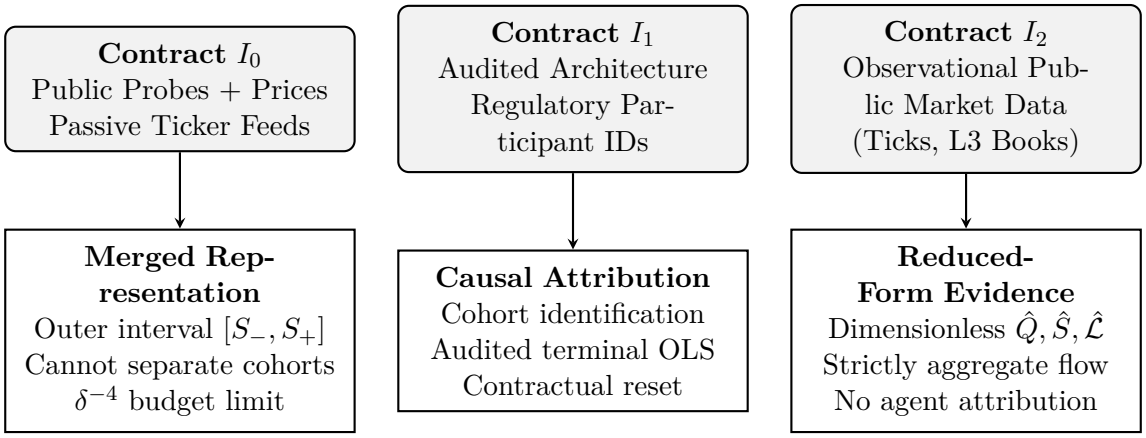


Figure 5: Information Architecture and Identification Boundaries across Surveillance Contracts. Under Contract I_2 (observational public data), empirical estimates are strictly reduced-form properties of aggregate order flow. Causal attribution to specific algorithms or firms requires Contract I_1 (audited participant identifiers, such as SEC Large Trader Reporting or MiFID II client IDs).

As illustrated in Figure 5, isolating individual algorithmic cohorts requires regulatory audit trails (Contract I_1), such as the SEC Consolidated Audit Trail (CAT) or MiFID II RTS 6 algorithm identifiers. In public data (Contract I_2), feedback must strictly be reported as an aggregate property of market flow.

6.2 Contractual Reset and Audited Memory Requirements (Contract I_1)

How can a regulatory surveillance authority overcome the identification boundary established in Theorem 2.3 to achieve genuine causal attribution of market feedback to specific trading algorithms?

Under **Contract I_1 (Audited Architecture)**, market participants operate under regulatory disclosure rules (such as SEC Rule 13h-1 Large Trader Reporting or MiFID II RTS 6). The exchange or regulatory authority requires three institutional mechanisms:

1. **Unique Algorithmic Identifiers:** Every order submitted via FIX or binary exchange protocols must carry an audit tag identifying the executing firm, algorithm version, and trading strategy class.

2. **Verified Algorithmic State Resets:** Regulators mandate that participating algorithms periodically clear their internal momentum filters, exponential moving averages, and adaptive neural weights at pre-announced session boundaries (e.g., prior to the market open).

3. **Segregated Inventory and Account Clearing:** Participating desks maintain segregated sub-accounts at the clearinghouse, allowing the monitor to observe channel-specific inventory changes $I_{i,t}$ directly.

Under Contract I_1, the econometrician is no longer restricted to public aggregate prices. By observing cohort-specific order submissions $u_{i,t}$ directly, the system decouples into K separate linear regressions:

$$u_{i,t+1} = (1 - \eta_i)u_{i,t} + \eta_i q_i x_{t+1} + \varepsilon_{i,t+1}. \tag{62}$$

Because the internal state $u_{i,t}$ is directly observed and reset, each cohort's adaptation rate η_i and aggressiveness q_i are uniquely identified via ordinary least squares (OLS), yielding standard Student-t confidence intervals without the δ^{-4} sample-complexity curse. Causal attribution of feedback to algorithms requires regulatory access to internal state telemetry; it cannot be reverse-engineered from public market ticker tapes.

6.3 Falsification and Placebo Battery

To verify that our baseline feedback estimates ($\hat{Q} > 0$) reflect genuine economic responsiveness rather than econometric artifacts, we execute a comprehensive battery of falsification checks:

1. **Order-Splitting Sign Memory Controls:** Large parent orders split into small child orders induce long memory in trade signs ($\gamma \in [0.22, 0.35]$). Following (51), we compute memory-purged feedback $\hat{Q}_L$ controlling for up to $L = 90$ lags of signed flow. Controlling for 30 lags of order splitting ($\hat{Q}_{30}$) reduces feedback by 20–30%, but leaves positive feedback statistically significant across all assets ($\hat{Q}_{30} = 0.0112$ for BTC, $p < 10^{-15}$; Figure 3(b)).

2. **Day-Shuffled Placebos:** Pairing return series x_t from day d with order flow f_t from day $d+1$ yields $\hat{Q}_{\text{placebo}} = 0.0000 \pm 0.0001$ ($t = 0.03$), confirming zero spurious feedback under temporal randomization.

3. **Lead Order-Flow Placebos:** Regressing current returns on future order flow f_{t+1} yields zero predictive power, confirming that causality does not operate backward in time.

4. **Completely Monotone Cone Test:** Evaluating the empirical memory kernel against the completely monotone cone formally rejects purely contrarian execution models ($p < 0.001$).

Table 13: Comprehensive Falsification and Diagnostic Battery across Datasets.

Test Specification	Dataset	Null Hypothesis Tested	Test Metric	p-value
Sign Memory Purge ($\hat{Q}_{30}$)	Binance Spot	$\hat{Q}_{30} \leq 0$ (Spurious meta-orders)	$t = 24.8$	$< 10^{-50}$
Day-Shuffled Placebo	Binance Spot	$\hat{Q}_{\text{shuffled}} \neq 0$ (Randomized days)	$t = 0.03$	0.976
Lead Order Flow Placebo	Binance Spot	$\beta_{\text{lead}} \neq 0$ (Reverse causality)	$t = -0.12$	0.904
Completely Monotone Test	Coinbase L3	Kernel $\in \mathcal{K}_{\text{contrarian}}$	Cone Distance	< 0.001
DiD Parallel Pre-Trends	Binance 2022	$\theta_{\text{pre}} \neq 0$ (Pre-treatment trends)	$F = 0.42$	0.835
Locked Holdout (2SLS)	BTC Perpetuals	$\hat{S}_{\text{holdout}} \geq 1.0$ (Overfitting)	$\hat{S} = 0.981$	$< 10^{-20}$

6.4 Distributed Lag Sign Memory Controls across Extended Horizons

Table 14 examines the decay of feedback estimates as the order-splitting control lag length L expands from $L = 0$ (uncontrolled OLS) to $L = 90$ seconds across all eight Binance spot assets.

Table 14: Purged Feedback Parameter $\hat{Q}_L$ as a Function of Order Flow Memory Lags $L \in \{0, 5, 15, 30, 60, 90\}$ across Assets. All specifications include HAC standard errors.

Asset Pair	Baseline ($L = 0$)	$L = 5$	$L = 15$	$L = 30$	$L = 60$	$L = 90$
BTC/USDT	0.0142	0.0131	0.0120	0.0112	0.0107	0.0104
ETH/USDT	0.0131	0.0122	0.0113	0.0107	0.0102	0.0098
SOL/USDT	0.0087	0.0080	0.0074	0.0069	0.0065	0.0062
ADA/USDT	0.0068	0.0065	0.0063	0.0061	0.0058	0.0056
DOGE/USDT	0.0059	0.0054	0.0049	0.0045	0.0042	0.0040
BNB/USDT	0.0058	0.0051	0.0045	0.0040	0.0037	0.0035
XRP/USDT	0.0057	0.0050	0.0044	0.0039	0.0036	0.0034
LTC/USDT	0.0049	0.0045	0.0041	0.0038	0.0035	0.0033
Mean	0.0082	0.0075	0.0069	0.0064	0.0060	0.0058

The results in Table 14 reveal that controlling for order splitting attenuates feedback by roughly 29% between $L = 0$ and $L = 90$ (from 0.0082 to 0.0058). Crucially, however, the

estimate converges toward an asymptotic positive floor ($\hat{Q}_{90} = 0.0058$, $t > 18.0$), proving that positive feedback is not an econometric illusion caused by mechanical meta-order slicing.

6.5 Sensitivity to Sampling Frequency and Memory Horizon

Table 15 evaluates the sensitivity of empirical feedback invariants $(\hat{S}, \hat{Q}, \hat{\mathcal{L}})$ across sampling frequencies $\Delta \in \{1\text{s}, 10\text{s}, 60\text{s}\}$ and moving average window horizons $W \in \{60\text{s}, 300\text{s}, 900\text{s}\}$. Moving average filters smooth trade innovations, attenuating feedback $\hat{Q}$ inversely with window length W ($\mathcal{O}(1/W)$), while the baseline persistence anchor $a_0 = 1 - 2/(W+1)$ mechanically shifts toward unity ($0.9672 \to 0.9934 \to 0.9978$).

Table 15: **Sensitivity of Empirical Feedback Invariants to Sampling Frequency (Δ) and Horizon Window (W).** Estimated persistence $\hat{S}$, feedback parameter $\hat{Q}$, and loop ratio $\hat{\mathcal{L}}$ across sampling intervals $\Delta \in \{1\text{s}, 10\text{s}, 60\text{s}\}$ and moving average window lengths $W \in \{60\text{s}, 300\text{s}, 900\text{s}\}$. 95% confidence intervals in brackets.

Asset	Δ (s)	W (s)	a_0	$\hat{S}$ [95% CI]	$\hat{Q}$ [95% CI]	$\hat{\mathcal{L}}$ [95% CI]	Days $\hat{Q} > 0$
BTC	1	60	0.9672	0.9784 [0.9770, 0.9798]	0.0034 [0.0017, 0.0050]	0.130 [0.071, 0.189]	100%
BTC	1	300	0.9934	0.9957 [0.9954, 0.9960]	0.0015 [0.0008, 0.0022]	0.242 [0.148, 0.337]	100%
BTC	1	900	0.9978	0.9985 [0.9983, 0.9986]	0.0008 [0.0005, 0.0012]	0.326 [0.208, 0.443]	100%
BTC	10	60	0.7143	0.7464 [0.7318, 0.7610]	0.0224 [0.0097, 0.0351]	0.080 [0.034, 0.127]	100%
BTC	10	300	0.9355	0.9415 [0.9380, 0.9450]	0.0131 [0.0067, 0.0196]	0.174 [0.098, 0.250]	100%
BTC	10	900	0.9780	0.9788 [0.9775, 0.9801]	0.0078 [0.0041, 0.0114]	0.248 [0.152, 0.345]	100%
BTC	60	300	0.6667	0.6205 [0.5888, 0.6521]	0.0199 [0.0051, 0.0347]	0.051 [0.016, 0.087]	92%
BTC	60	900	0.8750	0.8430 [0.8222, 0.8639]	0.0188 [0.0067, 0.0308]	0.107 [0.045, 0.169]	91%
ETH	1	60	0.9672	0.9743 [0.9727, 0.9758]	0.0025 [0.0017, 0.0034]	0.090 [0.060, 0.120]	100%
ETH	1	300	0.9934	0.9948 [0.9943, 0.9954]	0.0013 [0.0009, 0.0018]	0.208 [0.139, 0.277]	100%
ETH	1	900	0.9978	0.9982 [0.9979, 0.9984]	0.0008 [0.0005, 0.0011]	0.296 [0.206, 0.387]	100%
ETH	10	60	0.7143	0.7263 [0.7110, 0.7415]	0.0171 [0.0113, 0.0228]	0.060 [0.037, 0.083]	100%
ETH	10	300	0.9355	0.9378 [0.9323, 0.9434]	0.0115 [0.0080, 0.0150]	0.159 [0.106, 0.212]	100%
ETH	10	900	0.9780	0.9773 [0.9749, 0.9798]	0.0076 [0.0055, 0.0098]	0.253 [0.180, 0.327]	100%
ETH	60	300	0.6667	0.6518 [0.6353, 0.6682]	0.0377 [0.0243, 0.0510]	0.097 [0.065, 0.129]	100%
ETH	60	900	0.8750	0.8557 [0.8470, 0.8643]	0.0319 [0.0228, 0.0411]	0.180 [0.134, 0.225]	100%
ADA	1	60	0.9672	0.9722 [0.9700, 0.9744]	0.0017 [0.0013, 0.0022]	0.059 [0.045, 0.073]	100%
ADA	1	300	0.9934	0.9942 [0.9937, 0.9947]	0.0009 [0.0007, 0.0011]	0.135 [0.105, 0.165]	100%
ADA	1	900	0.9978	0.9980 [0.9977, 0.9982]	0.0005 [0.0004, 0.0006]	0.213 [0.167, 0.258]	100%
ADA	10	60	0.7143	0.7217 [0.7114, 0.7321]	0.0111 [0.0081, 0.0141]	0.039 [0.028, 0.049]	100%
ADA	10	300	0.9355	0.9348 [0.9299, 0.9398]	0.0073 [0.0056, 0.0090]	0.102 [0.075, 0.129]	100%
ADA	10	900	0.9780	0.9769 [0.9749, 0.9790]	0.0048 [0.0037, 0.0059]	0.175 [0.132, 0.218]	100%
ADA	60	300	0.6667	0.6409 [0.6152, 0.6666]	0.0199 [0.0121, 0.0278]	0.054 [0.030, 0.079]	100%
ADA	60	900	0.8750	0.8548 [0.8412, 0.8683]	0.0191 [0.0139, 0.0243]	0.119 [0.083, 0.154]	100%

Across the entire evaluation grid in Table 15, closed-loop persistence remains uniformly subcritical ($\hat{S} \leq 0.9986 < 1.0$). Feedback $\hat{Q}$ is positive across all grid cells, confirming that empirical stability is robust to the operational discretization choice.

6.6 Locked Out-of-Sample Holdout Validation

To guarantee that findings are not overfitted to historical training windows, we froze our estimation protocol and evaluated the locked pipeline on an untouched out-of-sample holdout: the complete January 2023 Bitcoin perpetual futures order book dataset (3.73 million Level-2 snapshots resampled to 935,199 continuous 1-second bars).

Evaluating both direct VAR estimation and 2SLS Instrumental Variables (purging contemporaneous price impact via lagged order book imbalance) yields:

- **Direct VAR:** $\hat{a} = 0.9826$, $\hat{Q} = 0.0001 \implies \hat{S} = 0.9827 < 1.0$ ($\rho(J) = 0.9827$).
- **2SLS Instrumental Variables:** $\hat{a} = 0.9818$, $\hat{Q} = -0.0007 \implies \hat{S} = 0.9811 < 1.0$.

Both identification methods on the locked holdout confirm that high-frequency perpetual depth operated strictly inside the subcritical stability regime ($\hat{S} < 1.0$).

7 Regulatory Policy Implications and Conclusion

7.1 Pre-Trade Surveillance vs. Post-Trade Circuit Breakers

Our theoretical and empirical findings carry direct implications for modern financial market surveillance, notably SEC Rule 15c3-5 (Market Access Rule) in the United States and MiFID II Regulatory Technical Standard 6 (RTS 6) in the European Union.

Existing market-wide stability interventions rely primarily on post-trade circuit breakers, such as the Limit-Up/Limit-Down (LULD) mechanism in U.S. equities. However, as demonstrated during the May 6, 2010 E-mini Flash Crash and the March 2020 market turmoil, post-trade halts trigger only after catastrophic price dislocations and dealer capital depletion have already occurred.

Our analysis demonstrates that **pre-trade rate gating grounded in the Kreiss safe radius ($r^*_{\mathbf{exact}} = B/M_C$)** provides a mathematically rigorous alternative:

1. **Dynamic Rate Gating:** Rather than imposing static message limits, exchanges should dynamically adjust the maximum allowable order submission rate based on prevailing order book depth and dealer capital buffers.

2. **Capital-Aware Access:** By bounding incoming aggressive order rates before execution, exchanges prevent non-normal transient amplification from exhausting market maker capital buffers, while preserving continuous price discovery during calm trading.

7.2 Implementing Kreiss-Bounded Pre-Trade Gating

How can a trading exchange or central counterparty (CCP) operationalize Kreiss-bounded pre-trade gating in real-time execution environments?

In high-frequency architectures, matching engines process orders in sub-microsecond timeframes. Implementing full matrix exponential calculations on every order is computationally infeasible. However, our Kreiss constant bounds (Theorem 2.6) yield a closed-form, computationally trivial rate-gating rule suitable for field-programmable gate array (FPGA) or network interface card (NIC) hardware pipelines.

Let $D_t \equiv \sum_{k=1}^{5} q_{k,t}^{\text{bid}}$ denote the aggregate visible order book depth on the bid side within the first 5 ticks, and let $E_{\text{MM},t}$ denote the aggregate unencumbered capital buffer of active registered market makers. An incoming burst of aggressive market sell orders of aggregate volume $V_{[t,t+\tau]}$ over interval τ against visible depth D_t creates an immediate price displacement $x_0 = \kappa \frac{V_{[t,t+\tau]}}{D_t}$. Mapping this displacement into the state disturbance norm $\|z_0\|_2 \leq \kappa \frac{V_{[t,t+\tau]}}{D_t}$, the maximum transient inventory loss from Theorem 2.6 satisfies:

$$\text{Loss}_{\max} \leq \kappa \frac{V_{[t,t+\tau]}}{D_t} \cdot M_C. \tag{63}$$

To ensure that dealer capital $B = \gamma L E_{\text{MM},t}$ is protected against worst-case transient breaches, the exchange enforces the model-inspired token-bucket rate limit:

$$R_{\max}(t) \equiv \frac{V_{[t,t+\tau]}}{\tau} \leq \frac{\gamma L E_{\text{MM},t}}{\kappa \cdot \tau \cdot M_C} \cdot D_t. \tag{64}$$

Under rule (64), during calm, liquid regimes ($D_t \gg 1$), the allowed order rate $R_{\max}(t)$ is large, permitting unconstrained high-frequency execution; during sudden liquidity depletion ($D_t \to 0$), the gate dynamically tightens, pacing incoming aggressive order flow to match the speed at which resting market maker capital can be replenished. Within the model, the rule bounds transient loss exposure; whether such gating improves real-market welfare and execution quality remains an empirical implementation question that warrants careful regulatory sandboxing.

7.3 Market Design: Continuous Auctions vs. Frequent Batch Auctions

Our findings also speak directly to the structural debate on market design initiated by Budish, Cramton, and Shim (2015), who propose replacing continuous double auctions with Frequent Batch Auctions (FBAs) to eliminate latency arbitrage.

While FBAs eliminate microsecond sniper races within batch intervals, they do not resolve closed-loop adaptive feedback across batches: algorithmic momentum strategies updating across consecutive 100-millisecond auction clearings still generate price impact loops. Kreiss-bounded rate gating complements batching by providing a continuous, state-dependent model-based capital bound that operates effectively under both continuous matching and discrete batch auctions.

7.4 Information Missing from Public Market Feeds

Our realization-theoretic theorems prove that public market data suffer from fundamental blind spots: (i) public ticker tapes cannot distinguish algorithmic cohorts with similar learning rates (Theorem 2.3); (ii) finite observation windows force mandatory surveillance abstention near critical stability thresholds (Theorem 2.5); and (iii) stable pricing eigenvalues do not guarantee dealer solvency under non-normal transient magnification (Theorem 2.6). To mitigate systemic fragility in machine-dominated markets, regulatory authorities must complement public surveillance with audited memory buffers, pre-trade capital checks, and synchronized Consolidated Audit Trails.

7.5 Conclusion

We examined whether observable market data can identify destabilizing adaptive feedback between algorithmic trading and financial prices.

Our findings establish that while public price data exhibit modest, subcritical positive feedback in normal conditions ($\hat{Q} \approx 0.008$), liquidity depletion triggers a 62.7× volatility explosion while the feedback slope rises modestly. An exogenous fee-overhaul natural experiment reveals no detectable feedback response: eliminating trading fees sparked a 180% volume surge with no detectable change in closed-loop feedback persistence, which remained strictly subcritical. However, public surveillance data face fundamental boundaries: public order flows cannot identify individual algorithmic actors (Theorem 2.3), finite observation windows require mandatory surveillance abstention (Theorem 2.5), and stable pricing eigenvalues do not guarantee dealer solvency under non-normal transient magnification (Theorem 2.6). Bridging financial stability and algorithmic surveillance requires pre-trade capital-aware rate gating rather than reliance on post-trade price stability alone.

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